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Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Non-Decomposing Ablating Hypersonic Thermal Protection Systems

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This work presents a *physics-infused reduced-order modeling* (PIROM) framework towards design, analysis, and optimization of non-decomposing ablating hypersonic thermal protection systems (TPS), and is demonstrated via the modeling of transient thermo-ablative responses of multi-layered hypersonic TPS. The PIROM architecture integrates a *reduced-physics model* (RPM) as the building block, which is based on the *lumped capacitance model* (LCM) coupled to a *surface recession model* (SRM). The RPM provides a low-fidelity estimate of the thermo-ablative response of the TPS, based on average temperatures and one-dimensional surface displacements. This RPM is extended with data-driven hidden dynamics that are formulated via a systematic coarse-graining approach rooted in the *Mori-Zwanzig* (MZ) formalism, and that are learned from high-fidelity simulation data. Therefore, while the LCM and SRM capture the dominant physics of the ablating TPS response, the correction terms compensate for residual dynamics arising from higher-order non-linear interactions and temperature-advection effects due to surface recession. The trained PIROM consistently achieves errors of $\approx 0.5\%$ for a wide range of extrapolative settings of design parameters involving time-and-space varying boundary conditions and SRM models, and improves by an order of magnitude by the LCM alone. Moreover, the PIROM delivers RPM-level computational costs, enabling evaluations that are two orders of magnitude faster than the high-fidelity full-order model (FOM). These results demonstrate that PIROM effectively reconciles the trade-offs between accuracy, generalizability, and efficiency, providing a promising framework for optimizing multi-physics dynamical systems, such as TPS, under diverse operating conditions.

Nomenclature

Roman Symbols

$\mathbf{A}, \mathbf{A}_i, \mathbf{B}, \mathbf{B}_{ij}$	System matrices and their block-wise components
c_p	Specific heat
$\mathcal{D}$	Dataset
$\mathbf{d}_m$	Mesh displacements
d	Dimension of the domain
E_i	The i-th element
$\mathcal{E}$	Collection of edges
e_{ij}	Boundary shared by elements i and j

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$\mathbf{f}, \mathbf{f}_i$	Forcing vector and the i-th component
$\mathcal{J}$	Objective function
$\mathbf{k}, k$	Thermal conductivity
ℓ	Cost function
M	Number of elements
$\mathcal{N}_i$	Collection of neighbors for element i
N	Number of layers in TPS
N_p	Number of steps in a trajectory
N_s	Number of high-fidelity trajectories
$\mathcal{P}, \mathcal{Q}$	Projection and Residual operators
P	Number of basis functions of an element
q	Heat flux
R	Thermal resistance
$\mathbf{r}^{(1)}, \mathbf{r}^{(2)}$	Resolved and Unresolved dynamics
T	Temperature
t, t_0, t_f	Time, initial time, final time
$\mathbf{u}, \mathbf{u}_i$	Full state vector and the states of element i
$\mathbf{v}$	Velocities
$\mathbf{w}, w_i$	One-dimensional displacements and its i-th component
x	Spatial coordinate
$\mathbf{y}$	State vector of PIROM
$\mathbf{z}$	Observables

Greek Symbols

$\boldsymbol{\beta}$	Hidden states
Γ	Domain boundary of partial differential equation
ϵ	Tolerance in training
$\mathbf{\Theta}$	Learnable parameters in machine learning model
$\boldsymbol{\kappa}, \tilde{\boldsymbol{\kappa}}$	Memory kernels
ρ	Density
σ	Penalty factor in Discontinuous Galerkin
$\mathbf{\Phi}, \mathbf{\Phi}^+$	Projection matrix and its pseudo-inverse
$\boldsymbol{\phi}_i, \phi_p^i$	Basis functions of element i, and its p-th component
$\boldsymbol{\varphi}_i^j$	Weight vector mapping states in element i to average temperature j

$\boldsymbol{\xi}$ Parameters for operating conditions

Ω Domain of partial differential equation

Other symbols

$\bar{\square}$ Coarse-grained quantity

$\square_{ij}$ Quantity related to elements i and j

$\square_{HF}$ Quantities computed using the high-fidelity solver

$\square_{q}$ Quantities related to Neumann boundary condition

$\square_{T}$ Quantities related to Dirichlet boundary condition

$\square_{m}$ Quantities related to the mesh

Acronyms

BC Boundary Condition

DG Discontinuous Galerkin

FEM Finite-Element Method

FOM Full-Order Model

IPG Interior-Penalty Galerkin

LCM Lumped-Capacitance Model

MZ Mori-Zwanzig

NN Neural Network

NODE Neural Ordinary Differential Equation

NRMSE Normalized Root-Mean Squared Error

OOD Out of Distribution

PIROM Physics-Infused Reduced-Order Modeling

ROM Reduced-Order Model

RPM Reduced-Physics Model

SRM Surface recession model

TPS Thermal Protection System

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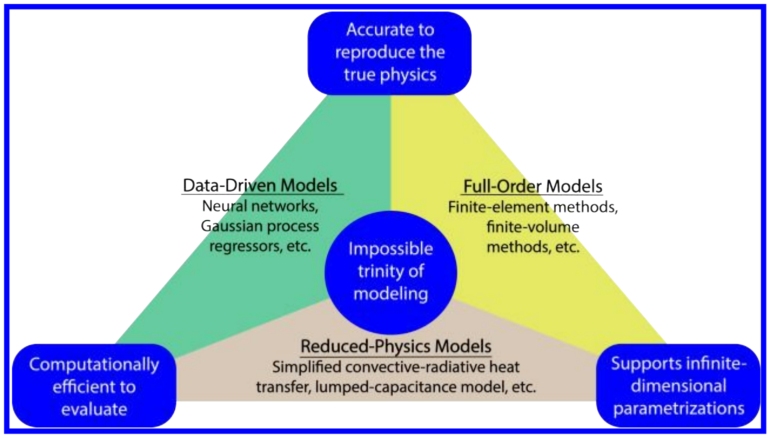

Fig. 1 The impossible trinity of modeling: accuracy, generalizability, and efficiency.

I. Introduction

At hypersonic speeds, aerospace vehicles experience extreme aero-thermo-dynamic environments that require specialized thermal protection systems (TPS) to shield internal sub-structures, electronics, and possibly crew members from the intense aerodynamic heating. The TPS is composed of ablating materials to withstand the high-energy physics – a high-temperature-capable and fibrous material, typically injected with a resin that fills the pore network and strengthens the composite [1]. The TPS design promotes the exchange of mass through thermal and chemical reactions (i.e., ablation), effectively mitigating heat transfer to the sub-structures. As a result, accurate prediction for the ablating TPS response under extreme hypersonic heating becomes critical to ensuring survivability, performance, and safety of hypersonic vehicles.

Even with today's advancements in computational resources and numerical methods, high-fidelity simulations of ablating TPS remains a formidable challenge, both theoretically and computationally. On the theoretical side, the thermo-chemical reactions, coupled with the irregular pore network structure and ablating boundaries, translate into complex non-linear equations governing multi-physical interactions across several spatio-temporal scales [1, 2]. On the computational side, numerical approaches based on finite-element (FEM) or finite-volume (FVM) methods yield systems of differential equations modeling the transient thermo-ablative response of the TPS [3]. The FEM discretizations lead to high-dimensional systems of equations, resulting in prohibitive computational costs for many-query applications such as design, optimization, uncertainty quantification, and real-time applications, where possibly thousands of model evaluations are required.

Reduced-order models (ROMs) have emerged as a promising approach to alleviate the computational costs of high-fidelity simulations [4, 5]. Ideally, a ROM should be: (1) accurate to reproduce high-fidelity solutions, (2) support continuous or infinite-dimensional design parameters such as geometrical shapes and material distributions, (3) be computationally efficient to evaluate to allow for fast turnaround times in design optimization. However, the above three capabilities usually form an *impossible trinity of modeling*, as illustrated in Fig. 1; building a ROM that achieves any two capabilities sacrifices the third.

The impossible trinity poses a significant challenge in the development of ROMs for the multi-disciplinary transient analysis and optimization of ablating TPS. Specifically, full-order models (FOMs), e.g., FEMs or FVMs, offer high accuracy and robust generalization over design spaces, but are computationally expensive to evaluate. Reduced-physics models (RPMs) – such as simplified convective-radiative heat transfer or engineering correlations – are low-dimensional models that achieve efficiency and broad applicability by ignoring higher-order non-linear effects. However, RPMs sacrifice accuracy for complex thermo-ablative responses due to the simplifications and assumptions inherent in their formulation, and it is generally not clear how to systematically leverage existing high-fidelity data to improve RPMs [6].

Lastly, data-driven ROMs, such as Gaussian Process Regression (GPR) [7], Neural Networks (NNs), and neural ordinary differential equations (NODEs) [8], can provide accurate and computationally-efficient approximations of high-fidelity models for complex thermo-ablative responses. However, these data-centric approaches often demand extensive high-fidelity data for training, do not necessarily satisfy fundamental physical constraints or conservation laws, and thus do not generalize well to the design spaces outside the training [9]. For example, our previous work demonstrated that NODEs trained on high-fidelity data of non-ablating TPS failed to generalize when subjected to

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boundary conditions and material models outside the training set [10].

This work presents the extension of the *physics-infused reduced-order modeling* (PIROM) framework to include effects of ablation for TPS applications, previously ignored in Ref. [10]. Specifically, the PIROM is demonstrated for the transient thermo-ablative response of multi-layered hypersonic TPS. The PIROM is a non-intrusive framework that combines the strengths of physics-based models with machine learning to formulate and train ROMs for parametrized non-linear dynamical systems. The backbone of the PIROM is the physics-based component, i.e., the RPM, which in this work is composed of: (1) a *lumped capacitance model* (LCM) to model the average heat transfer within the TPS layers, and (2) a *surface recession model* (SRM) to model one-dimensional surface ablation.

Leveraging the *Mori-Zwanzig* (MZ) formalism [11–13], the RPM is rigorously extended with data-driven hidden dynamics to account for the missing physics in the LCM, which are learned from high-fidelity data. The hidden dynamics enable higher predictive accuracy of the PIROM when subjected to complex boundary conditions and SRM model variations. For the TPS problem, the MZ approach produces a sufficiently simple model form while maintaining the physical consistency of the PIROM, as well as the dependence on design parameters. Thus, the PIROM aims to solve the impossible trinity of modeling by leveraging the generalizability and computational efficiency of RPMs, while incorporating the accuracy and adaptability of data-driven extensions. More importantly, the PIROM formulation provides a general methodology for developing PIROMs for other multi-physics problems.

The specific objectives of this work are summarized as follows:

1) Extend the previous PIROM formulation in Ref. [10] to model transient thermo-ablative response of multi-layered hypersonic TPS through a systematic coarse-graining procedure based on the Mori-Zwanzig formalism.
2) Benchmark the accuracy, generalizability, and computational accelerations of the PIROM against the RPM and the high-fidelity FEM solutions of ablating TPS, thus quantifying the PIROM's capabilities to solve the impossible trinity of modeling in complex multi-physical non-linear dynamical systems.

II. Modeling of Thermal Protection Systems

This section presents the problem of modeling the transient thermo-ablative response of a non-decomposing TPS, subjected to extreme hypersonic heating. Two different but mathematically connected solution strategies are provided: (1) a high-fidelity full-order model (FOM) based on a finite element method (FEM), and (2) a RPM based on a *lumped capacitance model* (LCM) coupled with a one-dimensional *surface recession model* (SRM). The FOM is computationally expensive but provides the highest fidelity, while the RPM is computationally efficient but has low predictive fidelity. However, both models are physically consistent to high-dimensional design variables. The following discussion presents the TPS modeling problem and the FOM and RPM solution strategies.

A. Governing Equations

The multi-physics of a non-decomposing ablating TPS under a hypersonic boundary layer involves the *energy equation* for heat conduction inside the TPS, and the *pseudo-elasticity equation* for mesh motion due to surface recession. The coupling between these two equations occurs at the heated boundary, where the surface temperature drives the surface recession velocity, which appears as an advection term in the energy equation. The governing PDEs are described as follows.

1. Energy Equation

Consider a generic domain $\Omega \subset \mathbb{R}^d$, $d = 2$ or 3, illustrated in Fig. 2. Let $\partial\Omega = \Gamma_q \cup \Gamma_T$ and $\Gamma_q \cap \Gamma_T = \emptyset$, where a Neumann $q_b(x,t)$ boundary condition is prescribed on the heated boundary Γ_q, and represents the surface exposed to the hypersonic boundary layer. The Dirichlet $T_b(x,t)$ boundary condition is prescribed on the boundary Γ_T. The TPS is divided into N non-overlapping components $\{\Omega_i\}_{i=1}^N$, as illustrated in Fig. 2 for $N = 2$. The i-th component Ω_i is associated with material properties $\left(\rho_i, c_{p,i}, \mathbf{k}_i\right)$, which are continuous within one component, and can be discontinuous across two neighboring components.

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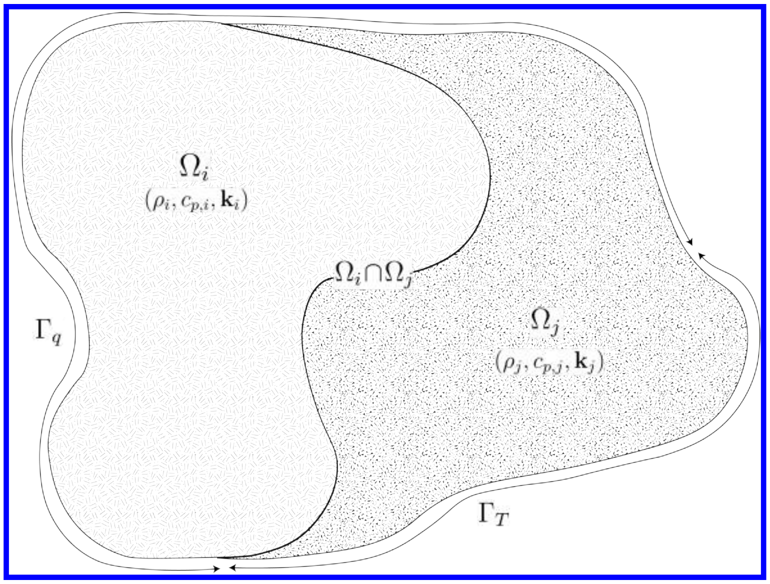

Fig. 2 General domain Ω with prescribed Neumann and Dirichlet boundary conditions on Γ_q and Γ_T. Mesh displacement $w(x,t)$ occurs on the Γ_q boundary.

Over the time domain $\mathcal{T} = [t_0, t_f]$, the energy equation describes the transient heat conduction inside the TPS,

$$\rho c_p \left(\frac{\partial T}{\partial t} + \tilde{\mathbf{v}}(x,t) \cdot \nabla T \right) - \nabla \cdot (\mathbf{k} \nabla T) = 0, \quad t \in \mathcal{T},\ x \in \Omega \tag{1a}$$

$$-\mathbf{k}\nabla T \cdot \mathbf{n} = q_b(x,t), \quad t \in \mathcal{T},\ x \in \Gamma_q \tag{1b}$$

$$T(x,t) = T_b(x,t), \quad t \in \mathcal{T},\ x \in \Gamma_T \tag{1c}$$

$$T(x,0) = T_0(x), \quad x \in \Omega \tag{1d}$$

where ρ, c_p, and $\mathbf{k} \in \mathbb{R}^{d \times d}$ are the constant density, heat capacity, and thermal conductivity, respectively. Note that our prior work has applied the PIROM to TPS problems with temperature-varying material properties [10]. In Eq. (1d) the $\rho c_p \frac{\partial T}{\partial t}$ term includes the unsteady energy storage, $\rho c_p \tilde{\mathbf{v}}(x,t) \cdot \nabla T$ includes the temperature advection due to ablation, and $\nabla \cdot (\mathbf{k}\nabla T)$ includes the heat conduction.

An Arbirtrary Lagrangian-Eulerian (ALE) description is used to account for mesh motion due to surface recession. The relative velocity of the material $\tilde{\mathbf{v}}(x,t)$ with respect to the mesh is,

$$\tilde{\mathbf{v}}(x,t) = \mathbf{v}_s(x,t) - \mathbf{v}_m(x,t) \tag{2}$$

where $\mathbf{v}_s(x,t)$ and $\mathbf{v}_m(x,t)$ are the physical material velocity and mesh velocity, respectively. In this work, the physical material velocity is assumed to be zero, i.e., $\mathbf{v}_s(x,t) = \mathbf{0}$, and thus the relative velocity is simply the negative of the mesh velocity, $\tilde{\mathbf{v}}(x,t) = -\mathbf{v}_m(x,t)$. The advection term in Eq. (1a) thus captures the effect of temperature advection due to mesh motion, and the governing equation for the mesh velocity is provided next.

2. Pseudo-Elasticity Equation

At time $t \in \mathcal{T}$, the mesh displacement field $\mathbf{d}_m \in \mathbb{R}^d$ is described by the steady-state pseudo-elasticity equation, which models the mesh as a fictitious elastic solid that deforms according to the prescribed Dirichlet boundary conditions. The mesh velocities are obtained by differentiating the mesh displacements with respect to time, i.e., $\mathbf{v}_m(x,t) = \dot{\mathbf{d}}_m$, which appear in the advection term of the energy equation in Eq. (1a) and is approximated using finite differences in the coupled simulation. The governing equation for mesh displacements is,

$$\nabla \cdot \boldsymbol{\sigma}(\mathbf{d}_m) = \mathbf{0}, \quad \forall\, x \in \Omega \tag{3a}$$

$$\mathbf{d}_m(x,t) = \mathbf{d}_q(x,t), \quad t \in \mathcal{T},\ \forall\, x \in \Gamma_q \tag{3b}$$

$$\mathbf{d}_m(x,t) = \mathbf{0}, \quad t \in \mathcal{T},\ \forall\, x \in \Gamma_T \tag{3c}$$

$$\mathbf{d}_m(x,t_0) = \mathbf{0}, \quad \forall\, x \in \Omega \tag{3d}$$

where the stress tensor $\boldsymbol{\sigma}$ is related to the strain tensor $\boldsymbol{\epsilon}(\mathbf{d}_m)$ through Hooke's law,

$$\boldsymbol{\sigma}(\mathbf{d}_m) = \mathbb{D} : \boldsymbol{\epsilon}(\mathbf{d}_m)$$

where $\mathbb{D}$ is the fourth-order positive definite elasticity tensor, and ":" is the double contraction of the full-order tensor $\mathbb{D}$ with the second-order tensor $\boldsymbol{\epsilon}$. In this work, the standard isotropic case is considered, where $\mathbb{D}$ is fully described by two Lamé parameters λ and μ arbitrarily. The symmetric strain tensor $\boldsymbol{\epsilon}$ measures the deformation of the mesh due to displacements $\mathbf{d}_m(x,t)$, and is defined as,

$$\boldsymbol{\epsilon}(\mathbf{d}_m) = \frac{1}{2}\left(\nabla \mathbf{d}_m + \nabla \mathbf{d}_m^\top\right)$$

The "material" properties for the mesh are chosen to tailor the mesh deformation, and need not represent the actual material being modeled [1].

The boundary conditions $\mathbf{d}_q$ and $\mathbf{d}_T = \mathbf{0}$ are Dirichlet-type on the heated Γ_q and unheated Γ_T boundaries, respectively, and the initial mesh displacement is set to zero as in Eq. (3d). The surface velocity due to ablation is a function of the surface temperature $T_q(x,t)$ for $x \in \Gamma_q$. For the i-th material component, the mesh velocity is imposed based on the following relation,

$$\hat{\mathbf{n}} \cdot \mathbf{v}_m(x,t) = f_i(T_q(x,t)), \quad x \in \Gamma_{q,i} \tag{4}$$

where $\Gamma_q = \cup_{i=1}^{\tilde{N}} \Gamma_{q,i}$ with $\Gamma_{q,i}$ as the portion of the heated boundary that belongs to the i-th ablative component, $\tilde{N}$ is the number of ablative components with $\tilde{N} \leq N$, $\hat{\mathbf{n}}$ is the unit normal vector, and f_i is a material-dependent function obtained from tabulated data, commonly referred to as a B' table [1]. The B' table provides a model for the recession velocity as a function of the surface temperature, and is pre-computed based on high-fidelity simulations or physical experiments for a one-dimensional slab of materials, and is independent of the TPS geometry. Provided the surface velocity, the boundary condition in Eq. (5) for the mesh displacements are computed by integrating the surface velocity over time,

$$\mathbf{d}_q(x,t) = \int_0^t \mathbf{v}_m(x,\tau)\, d\tau, \quad x \in \Gamma_q \tag{5}$$

B. Full-Order Model: Finite-Element Method

The transient thermo-ablative response of the TPS is modeled using a high-fidelity FOM based on FEM. The governing PDEs – namely, the energy equation in Eq. (1d) and the pseudo-elasticity equation in Eq. (3d) – are numerically solved using the SIERRA/Aria multi-physics code developed at Sandia National Laboratories [14]. While the standard practice in multi-dimensional ablative TPS solvers is to employ continuous Galerkin FEM for both the energy and mesh-motion equations, this work adopts a *discontinuous Galerkin* FEM (DG-FEM) formulation for the energy equation. The DG-FEM formulation is mainly for theoretical convenience to support the model-order reduction procedures in the subsequent PIROM derivations; equivalence between DG and standard FEM is noted upon their convergence. The following discussion presents the numerical solution of both governing equations, as well as the coupling scheme between them.

Energy Equation Consider a conforming mesh partition of the TPS domain into M elements $\{E_i\}_{i=1}^M$, where each element belongs exactly to one TPS component Ω_j. Let e_{ij} denote the interface between neighboring elements E_i and E_j, and let e_{iq} and e_{iT} denote the portions of the element boundary on the Neumann Γ_q and Dirichlet Γ_T boundaries, respectively. The quantity $|e|$ denotes the length ($d = 2$) or area ($d = 3$) of a component boundary e. For the i-th

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element, use a set of P trial functions, such as polynomials, to represent the temperature distribution,

$$T_i(x,t) = \sum_{l=1}^{P} \phi_l^i(x) u_l^i(t) \equiv \boldsymbol{\phi}_i^\top(x)\mathbf{u}_i(t), \quad i = 1, 2, \ldots, M \tag{6}$$

Applying the DG variational formulation to the energy equation, incorporating the numerical fluxes across all element interfaces, and assembling over the mesh yields the semi-discrete system of ODEs,

$$\mathbf{A}\dot{\mathbf{u}} = [\mathbf{B} + \mathbf{C}(\mathbf{u})]\,\mathbf{u} + \mathbf{f}(t) \tag{7}$$

where $\mathbf{u} = [\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_M]^\top \in \mathbb{R}^{MP}$ includes all the DG variables, $\mathbf{f} \in \mathbb{R}^{MP}$ is the external forcing, and the system matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are the matrices due to heat capacity, heat conduction, and temperature advection due to mesh motion, respectively. Note that the advection matrix $\mathbf{C}(\mathbf{u})$ is a function of the mesh velocity $\mathbf{v}_m$, which in turn is a function of temperature through the boundary condition in Eq. (4); this is the main source of non-linearity in the current TPS problem. A detailed derivation of Eq. (7) is provided in Appendix A.A.

Pseudo-Elasticity Equation The *steady pseudo-elasticity equation* is spatially discretized using the standard Galerkin FEM method on a structured mesh with quadrilateral elements. Define the scalar basis functions $\{\psi_j(x)\}_{j=1}^{Q}$ associated with the global displacement degrees of freedom $\{\mathbf{d}_j\}_{j=1}^{Q}$ where Q is the total number of nodes in the mesh. At time t, the mesh displacement field $\mathbf{d}_m(x,t)$ is approximated as,

$$\mathbf{d}_m(x,t) \approx \sum_{j=1}^{Q} \psi_j(x)\mathbf{d}_j \tag{8}$$

Substituting into the weak form of Eq. (3a), the following linear system of equations is obtained for the nodal displacements at each time step,

$$\mathbf{K}\mathbf{d} = \mathbf{g} \tag{9}$$

where $\mathbf{d}$ is the global displacement vector, $\mathbf{K} \in \mathbb{R}^{dQ\times dQ}$ the global stiffness matrix for a d-dimensional problem, defined by the volume integrals over the domain Ω provided the elasticity tensor $\mathbb{D}$, and $\mathbf{g}$ is the global force vector due to the Dirichlet boundary conditions on the heated Γ_q and unheated Γ_T boundaries.

Coupled Full-Order Model The temperature-dependent mesh motion on the heated boundary Γ_q establishes a two-way coupling scheme between the energy and pseudo-elasticity FEM models. At each time step, the coupling proceeds as follows: (1) *boundary displacement update*, (2) *mesh displacement solve*, and (3) *thermal solve with mesh-induced advection*. In the boundary displacement update, the surface temperature $T_q(x,t)$ on the heated boundary Γ_q is used to evaluate the recession velocity through the B' table as in Eq. (4). Integrating the velocity over the time step as in Eq. (4) updates the prescribed boundary displacement $\mathbf{d}_q(x,t)$ in Eq. (5). In the mesh displacement solve, the pseudo-elasticity equation in Eq. (9) is solved to obtain the mesh displacement field $\mathbf{d}_m(x,t)$ based on the updated boundary displacements. A finite-difference approximation is then applied to compute the mesh velocity $\mathbf{v}_m(x,t) = \dot{\mathbf{d}}_m$ based on history of displacement fields. In the thermal FEM solve with mesh-induced advection, the mesh velocity field enters the advection operator $\mathbf{C}(\mathbf{u})$ in the semi-discrete energy equation in Eq. (7). The energy equation is then advanced using an implicit-time integration scheme until the final time t_f is reached.

C. Reduced-Physics Model

The RPM for predicting the transient thermo-ablative response of the TPS consists of two components: (1) *surface recession model* (SRM) and a *lumped capacitance model* (LCM). The SRM provides a relation between the surface temperature and *one-dimensional* surface recession velocity based on pre-computed B' tables for the material, enabling the computation of *one-dimensional* surface displacements. Provided the geometry changed induced by the surface recession, the LCM predicts the average temperature inside each component of the TPS, which are in turn used as low-fidelity estimates for the surface temperatures required by the SRM. The SRM and LCM are coupled to define the RPM, providing low-fidelity estimates for temperatures and surface recessions of the ablating TPS.

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1. Surface Recession Model

The mesh displacements $\mathbf{d}_m$ are constrained to be *one-dimensional* on the heated boundary Γ_q, i.e., $w_i(x,t) = \mathbf{d}_m(x,t)\cdot\hat{\mathbf{n}}_i$, where $\hat{\mathbf{n}}_i$ is the unit normal vector on the heated boundary $\Gamma_{q,i}$. Let $\mathbf{w} = \left[w_1, w_2, \dots, w_{\tilde{N}}\right]^\top \in \mathbb{R}^{\tilde{N}}$ include the one-dimensional displacements for the $\tilde{N}$ ablating components on the heated boundary, where $\tilde{N} \le N$. The SRM is described as,

$$\dot{\mathbf{w}} = \mathbf{\Xi}\mathbf{u} - \tilde{\mathbf{f}} \tag{10}$$

where $\mathbf{\Xi} = \operatorname{diag}\left(\alpha_1, \dots, \alpha_{\tilde{N}}\right)$ and $\tilde{\mathbf{f}} = \left(\alpha_1 u_{0,1}, \dots, \alpha_{\tilde{N}} u_{0,\tilde{N}}\right)^\top$. The constants α_i are small material-dependent parameters, determined from the B' table, and $u_{0,i}$ is the constant initial temperature of the ablative component. The SRM provides a relation between the surface's temperature and recession velocity, based on pre-computed B' tables for the material.

2. Lumped Capacitance Model

A general form of the LCM is provided in this section; details regarding the derivation for the four-component TPS used in the results section are provided in Appendix A.C. Let Ω be partitioned into N non-overlapping components $\{\Omega_i\}_{i=1}^N$, as illustrated in Fig. 2 for $N = 2$. The domain Ω is a function of the surface displacements $\mathbf{w}$, and thus the geometry of each component Ω_i is time-varying in the LCM formulation. The LCM predicts the temporal variation of average temperatures in multiple shape-varying interconnected components [15]. From a point of view of energy conservation, the LCM leads to the following system of first-order ODEs for the average temperatures in the components,

$$\bar{\mathbf{A}}(\mathbf{w})\dot{\bar{\mathbf{u}}} = \bar{\mathbf{B}}(\mathbf{w})\bar{\mathbf{u}} + \bar{\mathbf{f}}(t) \tag{11}$$

Where the states and inputs,

$$\bar{\mathbf{u}} = [\bar{u}_1, \bar{u}_2, \dots, \bar{u}_N]^\top \in \mathbb{R}^N, \quad \bar{\mathbf{f}} = \left[\bar{f}_1, \bar{f}_2, \dots, \bar{f}_N\right]^\top \in \mathbb{R}^N \tag{12}$$

include the average temperatures $\bar{\mathbf{u}}$ and spatially-integrated inputs $\bar{\mathbf{f}}$ for the N components. For $i, j = 1, 2, \dots, N$, the (i,j)-th elements of the $\bar{\mathbf{A}} \in \mathbb{R}^{N\times N}$, $\bar{\mathbf{B}} \in \mathbb{R}^{N\times N}$, and $\bar{\mathbf{f}} \in \mathbb{R}^N$ matrices are given by,

$$\bar{A}_i = \begin{cases} \int_{\Omega_i} \rho c_p d\Omega_i, & i = j \\ 0, & i \neq j \end{cases}, \quad \bar{B}_{ij} = \begin{cases} \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \bar{B}^i_{ij}, & i = j \\ \bar{B}^{(j)}_{ij}, & i \neq j \end{cases}, \tag{13a}$$

$$\mathbf{f}_i = \begin{cases} |e_{iq}|\bar{q}_i + \frac{|e_{iT}|}{R_i}\bar{T}_i, & i = j \\ 0, & i \neq j \end{cases} \tag{13b}$$

where,

$$\bar{q}_i = \frac{1}{|e_{iq}|}\int_{e_{iq}} q_b de_{iq}, \quad \bar{T}_i = \frac{1}{|e_{iT}|}\int_{e_{iT}} T_b de_{iT}, \quad \bar{B}^i_{ij} = -\frac{|e_{ij}|}{R_{ij}}, \quad \bar{B}^j_{ij} = \frac{|e_{ij}|}{R_{ij}} \tag{14}$$

where R_{ij} is the equivalent thermal resistance between two neighboring components Ω_i and Ω_j, and R_i is the thermal resistance between component Ω_i and the Dirichlet boundary. Note that the heat capacitances and thermal resistances are computed based on the current geometry of each component, which is a function of $\mathbf{w}$ provided by the SRM.

3. Coupled Reduced-Physics Model

The SRM and LCM are combined to define the RPM for predicting the thermo-ablative response of the TPS under hypersonic boundary layers. Specifically, the RPM is defined as the LCM as in Eq. (11), where the *geometry-dependent* matrices $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ are updated at each time step based on the current displacements $\mathbf{w}$ provided by the SRM. The RPM is formally stated as,

$$\tilde{\mathbf{A}}(\mathbf{s})\dot{\mathbf{s}} = \tilde{\mathbf{B}}(\mathbf{s})\mathbf{s} + \tilde{\mathbf{F}}(t) \tag{15a}$$

$$\tilde{\mathbf{z}} = \mathbf{s} \tag{15b}$$

where the state $\mathbf{s} = [\bar{\mathbf{u}}, \mathbf{w}]^\top \in \mathbb{R}^{N+\tilde{N}}$ includes the *average temperature* and *one-dimensional surface displacements*, and $\tilde{N}$ is the number of ablating components with $\tilde{N} \le N$. Moreover, the observables are defined as $\mathbf{z} = [\bar{\mathbf{u}}, \mathbf{w}]^\top \in \mathbb{R}^{N+\tilde{N}}$.

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The matrices are given as,

$$\tilde{\mathbf{A}}(\mathbf{s}) = \begin{bmatrix} \bar{\mathbf{A}}(\mathbf{s}) & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}, \quad \tilde{\mathbf{B}}(\mathbf{s}) = \begin{bmatrix} \bar{\mathbf{B}}(\mathbf{s}) & \mathbf{0} \\ \mathbf{\Xi} & \mathbf{0} \end{bmatrix}, \quad \tilde{\mathbf{F}}(t) = \begin{bmatrix} \bar{\mathbf{f}}(t) \\ -\tilde{\mathbf{f}} \end{bmatrix} \tag{16}$$

In the matrices $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$, the surface displacements $\mathbf{w}$ are used to define the dimensions for the Ω_i component used in Eqs. (13b) and (14), thus effectively coupling the LCM and SRM.

D. Summary of Modeling Approaches

The FOM (i.e., FEM) and RPM (i.e., LCM with SRM) are two different but mathematically connected solution strategies. Particularly, the LCM in Eq. (11) not only resembles the functional form of the DG model in Eq. (7), but can be viewed as a special case of the latter, where the mesh partition is extremely coarse, and the trial and test functions are piece-wise constants. This removes all spatial variations within each component, and neglects advection effects due to mesh motion.

For example, consider the case where each component Ω_i is treated as one single element, and each element employs one constant basis function $\phi_i = 1$. The DG-FEM model for the i-th component simplifies to the scalar ODE,

$$\mathbf{A}^i = \bar{A}_i, \quad \mathbf{C}^i = 0, \quad \mathbf{B}^i_{ij} = -\sigma|e_{ij}|, \quad \mathbf{B}^j_{ij} = \sigma|e_{ij}|, \quad \mathbf{f}_i = |e_{iq}|\bar{q}_i + \sigma|e_{iT}|\bar{T}_i \tag{17}$$

Clearly, the LCM is a coarse zeroth-order DG model with the inverse of thermal resistance chosen as the element-wise penalty factors. Or conversely, the DG model is a refined version of LCM via *hp*-adaptation.

The FOM and RPM represent two extremes in the modeling fidelity and computational cost spectrum. On one hand, the FOM is the most accurate but computationally expensive to evaluate due to the fine mesh discretizations for both the temperature and displacement fields, leading to possibly millions of state variables. On the other hand, the RPM considers only the average temperature of the material, from which the displacements are obtained by integrating the velocity. The coarsened representation of the temperature field significantly reduces the number of state variables to only a few per component, and thus reducing the computational cost. However, this sacrifices local temperature information that becomes critical to properly capture higher-order effects due to mesh motion and thermal gradients within each component. Thus, neither the FOM nor the RPM is an universal approach for real-world analysis, design, and optimization tasks for ablating TPS, where thousands of high-fidelity model evaluations may be necessary. This issue motivates the development of the PIROM, which can achieve the fidelity of FOM at a computational cost close to the RPM, while maintaining the generalizability to model parameters.

III. Physics-Infused Reduced-Order Modeling

The formulation of PIROM for ablating TPS starts by connecting the FOM, i.e., the DG-FEM, and the RPM, i.e., the LCM, via a coarse-graining procedure. This procedure pinpoints the missing dynamics in the LCM when compared to DG-FEM. Subsequently, the Mori-Zwanzig (MZ) formalism is employed to determine the model form for the missing dynamics in PIROM. Lastly, the data-driven identification of the missing dynamics in PIROM is presented.

A. Deriving the Reduced-Physics Model via Coarse-Graining

The subsequent coarse-graining formulation is performed on the DG-FEM in Eq. (7) to derive the LCM in Eq. (11). This process constrains the trial function space of a full-order DG model to a subset of piece-wise constants, so that the variables $\mathbf{u}$, matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, and forcing vector $\mathbf{f}$ are all approximated using a single state associated to the average temperature. Note that the coarse-graining is exclusively performed on the thermal dynamics, as it is the surface temperature that drives the one-dimensional recession via the SRM. Hence, the coarse-graining of the mesh dynamics is not included in the following procedure.

1. Coarse-Graining of States

Consider a DG model as in Eq. (7) for M elements and an LCM as in Eq. (11) for N components; clearly $M \gg N$. Let $\mathcal{V}_j = \left\{ i \middle| E_i \in \Omega_j \right\}$ be the indices of the elements belonging to the j-th component, so $E_i \in \Omega_j$ for all $i \in \mathcal{V}_j$. The number of elements in the j-th component is $|\mathcal{V}_j|$. The average temperature on Ω_j is,

$$\bar{u}_j = \frac{1}{|\Omega_j|}\sum_{i\in\mathcal{V}_j}\int_{E_i}\boldsymbol{\phi}^{(i)}(x)^T\mathbf{u}^{(i)}d\Omega = \frac{1}{\left|\Omega_j\right|}\sum_{i\in\mathcal{V}_j}|E_i|\,\boldsymbol{\varphi}_i^{j\top}\mathbf{u}^{(i)},\quad j = 1,2,\dots,N \tag{18}$$

where $\left|\Omega_j\right|$ and $|E_i|$ denote the area ($d = 2$) or volume ($d = 3$) of component j and element i, respectively. The orthogonal basis functions are defined as $\boldsymbol{\varphi}_i^{j\top} = [1, 0, \dots, 0]^\top \in \mathbb{R}^P$.

Conversely, given the average temperatures of the N components, $\bar{\mathbf{u}}$, the states of an arbitrary element E_i is written as,

$$\mathbf{u}^{(i)} = \sum_{k=1}^{N}\boldsymbol{\varphi}_i^k\bar{u}_k + \delta\mathbf{u}^{(i)},\quad i = 1,2,\dots,M \tag{19}$$

where $\boldsymbol{\varphi}_i^k = 0$ if $i \notin \mathcal{V}_k$, and $\delta\mathbf{u}^{(i)}$ represents the deviation from the average temperature and satisfies the orthogonality condition $\boldsymbol{\varphi}_i^{k\top}\delta\mathbf{u}^{(i)} = 0$ for all k.

Equations Eqs. (18) and (19) are combined and written in matrix form as,

$$\bar{\mathbf{u}} = \boldsymbol{\Phi}^+\mathbf{u},\quad \mathbf{u} = \boldsymbol{\Phi}\bar{\mathbf{u}} + \delta\mathbf{u} \tag{20}$$

where $\boldsymbol{\Phi} \in \mathbb{R}^{MP\times N}$ is a matrix of $M \times N$ blocks, with the (i, j)-th block as $\boldsymbol{\varphi}_i^j$, $\boldsymbol{\Phi}^+ \in \mathbb{R}^{N\times MP}$ is the left inverse of $\boldsymbol{\Phi}$, with the (i, j)-th block as $\boldsymbol{\varphi}_i^{j+} = \frac{|E_i|}{|\Omega_j|}\boldsymbol{\varphi}_i^{j\top}$, and $\delta\mathbf{u}$ is the collection of deviations. By their definitions, $\boldsymbol{\Phi}^+\boldsymbol{\Phi} = \mathbf{I}$ and $\boldsymbol{\Phi}^+\delta\mathbf{u} = \mathbf{0}$.

2. Coarse-Graining of Dynamics

The dependence of the matrices with respect to the displacements $\mathbf{w}$ is dropped to isolate the analysis based on coarsened variables. Consider a function of states in the form of $\mathbf{M}(\mathbf{u})\,\mathbf{g}(\mathbf{u})$, where $\mathbf{g} : \mathbb{R}^{MP} \to \mathbb{R}^{MP}$ is a vector-valued function, and $\mathbf{M} : \mathbb{R}^{MP} \to \mathbb{R}^{p\times MP}$ is a matrix-valued function with an arbitrary dimension p. Define the projection matrix $\mathbf{P} = \boldsymbol{\Phi}\boldsymbol{\Phi}^+$ and the projection operator $\mathcal{P}$ as,

$$\begin{aligned}\mathcal{P}\left[\mathbf{M}(\mathbf{u})\mathbf{g}(\mathbf{u})\right] &= \mathbf{M}\left(\mathbf{P}\mathbf{u}\right)\mathbf{g}\left(\mathbf{P}\mathbf{u}\right)\\ &= \mathbf{M}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\mathbf{g}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\end{aligned} \tag{21}$$

so that the resulting function depends only on the average temperatures $\bar{\mathbf{u}}$. Correspondingly, the residual operator $\mathcal{Q} = \mathcal{I} - \mathcal{P}$, and $\mathcal{Q}\left[\mathbf{M}(\mathbf{u})\mathbf{g}(\mathbf{u})\right] = \mathbf{M}(\mathbf{u})\mathbf{g}(\mathbf{u}) - \mathbf{M}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\mathbf{g}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)$. When the function is not separable, the projection operator is simply defined as $\mathcal{P}\left[\mathbf{g}(\mathbf{u})\right] = \mathbf{g}\left(\mathbf{P}\mathbf{u}\right)$.

Subsequently, the operators defined above are applied to coarse-grain the dynamics. First, write the DG-FEM in Eq. (7) as,

$$\dot{\mathbf{u}} = \mathbf{A}^{-1}\mathbf{B}\mathbf{u} + \mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} + \mathbf{A}^{-1}\mathbf{f}(t) \tag{22}$$

and multiply both sides by $\boldsymbol{\Phi}^+$ to obtain,

$$\boldsymbol{\Phi}^+\dot{\mathbf{u}} = \boldsymbol{\Phi}^+\left(\boldsymbol{\Phi}\dot{\bar{\mathbf{u}}} + \delta\dot{\mathbf{u}}\right) = \dot{\bar{\mathbf{u}}} = \boldsymbol{\Phi}^+\mathbf{r}(\mathbf{u}, t) \tag{23}$$

Apply the projection operator $\mathcal{P}$ and the residual operator $\mathcal{Q}$ to the right-hand side to obtain,

$$\dot{\bar{\mathbf{u}}} = \mathcal{P}\left[\boldsymbol{\Phi}^+\mathbf{r}(\mathbf{u}, t)\right] + \mathcal{Q}\left[\boldsymbol{\Phi}^+\mathbf{r}(\mathbf{u}, t)\right] \equiv \mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) + \mathbf{r}^{(2)}(\mathbf{u}, t) \tag{24}$$

where $\mathbf{r}^{(1)}(\bar{\mathbf{u}}, t)$ is resolved dynamics that depends on $\bar{\mathbf{u}}$ only, and $\mathbf{r}^{(2)}(\mathbf{u}, t)$ is the un-resolved or residual dynamics. Detailed derivations and analysis of $\mathbf{r}^{(1)}(\bar{\mathbf{u}}, t)$ and $\mathbf{r}^{(2)}(\mathbf{u}, t)$ can be found in Appendix A.B.

It follows from our previous work in Ref. [10] that the resolved dynamics is exactly the LCM, where the advection term reduces to zero, i.e., $\bar{\mathbf{C}}(\bar{\mathbf{u}}) = \mathbf{0}$, as shown in Appendix A.B. Using the notation from Eq. (11), it follows that,

$$\begin{aligned}\mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) &= \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{B}}(\mathbf{w})\bar{\mathbf{u}} + \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{C}}(\bar{\mathbf{u}})\bar{\mathbf{u}} + \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{f}}(\bar{\mathbf{u}})\\ &= \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{B}}(\mathbf{w})\bar{\mathbf{u}} + \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{f}}(t)\end{aligned} \tag{25}$$

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where the following relations hold,

$$\bar{\mathbf{A}}(\mathbf{w}) = \mathbf{W}\left(\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{\Phi}\right)^{-1} \qquad \bar{\mathbf{C}}(\bar{\mathbf{u}}) = \mathbf{0} \tag{26a}$$

$$\bar{\mathbf{B}}(\mathbf{w}) = \mathbf{W}\mathbf{\Phi}^{+}\mathbf{B}\mathbf{\Phi} \qquad \bar{\mathbf{f}}(t) = \mathbf{W}\mathbf{\Phi}^{+}\mathbf{f} \tag{26b}$$

where $\mathbf{W} \in \mathbb{R}^{N\times N}$ is a diagonal matrix with the i-th element as $[\mathbf{W}]_{ii} = |\mathcal{V}_k|$ if $i \in \mathcal{V}_k$. The examination of the second residual term $\mathbf{r}^{(2)}(\mathbf{u}, t)$ in Eq. (24) is shown in the Appendix, and demonstrates that the physical sources of missing dynamics in the LCM including the approximation of non-uniform temperature within each component as a constant, and the elimination of the advection term due to coarse-graining. In sum, Eq. (25) shows that the LCM is a result of coarse-graining of the full-order DG-FEM, and reveals the sources of discrepancy between the LCM and the full-order DG-FEM. These discrepancies propagate into the SRM, which, as a result of the averaging in LCM, under-predicts the surface recession rates. In the subsequent section, the discrepancies in the LCM are corrected for via physics infusion.

B. Physics-Infusion Via Mori-Zwanzig Formalism

The Mori-Zwanzig (MZ) formalism is an operator-projection technique used to derive ROMs for high-dimensional dynamical systems, especially in statistical mechanics and fluid dynamics [11–13]. It provides an exact reformulation of a high-dimensional Markovian dynamical system, into a low-dimensional observable non-Markovian dynamical system. The proposed ROM is subsequently developed based on the approximation to the non-Markovian term in the observable dynamics. Particularly, Eq. (24) shows that the DG-FEM dynamics can be decomposed into the resolved dynamics $\mathbf{r}^{(1)}(\bar{\mathbf{u}}, t)$ and the orthogonal dynamics $\mathbf{r}^{(2)}(\mathbf{u}, t)$, in the sense of $\mathcal{P}\mathbf{r}^{(2)} = 0$. In this case, the MZ formalism can be invoked to express the dynamics $\bar{\mathbf{u}}$ in terms of $\bar{\mathbf{u}}$ alone as the projected Generalized Langevin Equation (GLE) [11–13],

$$\dot{\bar{\mathbf{u}}}(t) = \mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) + \int_0^t \tilde{\boldsymbol{\kappa}}(t, s, \bar{\mathbf{u}})ds \tag{27}$$

where the first and second terms are referred to as the Markovian and non-Markovian terms, respectively. The non-Markovian term accounts for the effects of past un-resolved states on the current resolved states via a memory kernel $\tilde{\boldsymbol{\kappa}}(t, s, \bar{\mathbf{u}})$, which in practice is computationally expensive to evaluate.

1. Markovian Reformulation

This section details the formal derivation of the PIROM as a system of ODEs for the thermal dynamics, based on approximations to the memory kernel. Specifically, the kernel $\tilde{\boldsymbol{\kappa}}$ is examined via a leading-order expansion, based on prior work [16]; this can be viewed as an analog of zeroth-order holding in linear system theory with a sufficiently small time step. In this case, the memory kernel is approximated as,

$$\tilde{\boldsymbol{\kappa}}(t, s, \bar{\mathbf{u}}) \approx \mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) \cdot \nabla_{\bar{\mathbf{u}}}\mathbf{r}^{(2)}(\mathbf{\Phi}\bar{\mathbf{u}}, t) \tag{28}$$

Note that the terms in $\mathbf{r}^{(1)}$ have a common factor $\bar{\mathbf{A}}^{-1}$; this motivates the following heuristic modification of the model form in Eq. (27),

$$\dot{\bar{\mathbf{u}}} = \mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) + \bar{\mathbf{A}}(\mathbf{w})^{-1}\int_0^t \boldsymbol{\kappa}(t, s, \bar{\mathbf{u}})ds \tag{29a}$$

$$\bar{\mathbf{A}}(\mathbf{w})\dot{\bar{\mathbf{u}}} = \bar{\mathbf{B}}(\mathbf{w})\bar{\mathbf{u}} + \bar{\mathbf{f}}(t) + \int_0^t \boldsymbol{\kappa}(t, s, \bar{\mathbf{u}})ds \tag{29b}$$

where the original kernel $\tilde{\boldsymbol{\kappa}}$ is effectively normalized by $\bar{\mathbf{A}}(\mathbf{w})^{-1}$. Intuitively, such choice of kernel reduces its dependency on the averaged material properties, and simplifies the subsequent design of model form.

Subsequently, the hidden states are introduced to "Markovianize" the system Eq. (27). In this manner, Eq. (29b) is converted into a pure state-space model, with the functional form of the LCM retained; since LCM is a physics-based model, then it encodes the physical information and retains explicit parametric dependence of the problem. Consider

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the representation of the kernel as a finite sum of simpler functions, e.g., exponentials,

$$\boldsymbol{\kappa}(t,s,\bar{\mathbf{u}}) = \sum_{j=1}^{m} \mathcal{K}_j(t,s,\bar{\mathbf{u}}) \left[\mathbf{p}_j + \mathbf{d}_j(\bar{\mathbf{u}})\right] \phi_j(s,\bar{\mathbf{u}}) \tag{30}$$

where,

$$\mathcal{K}_j(t,s,\bar{\mathbf{u}}) = e^{-\int_s^t (\lambda_j + e_j(\bar{\mathbf{u}}))d\tau}, \quad \phi_j(s,\bar{\mathbf{u}}) = \mathbf{q}_j^\top \bar{\mathbf{u}}(s) + \mathbf{g}_j(\bar{\mathbf{u}})^\top \bar{\mathbf{u}}(s) + \mathbf{r}_j^\top \bar{\mathbf{f}}(s) \tag{31}$$

with suitable coefficients $\mathbf{p}_j, \mathbf{d}_j, \mathbf{q}_j, \mathbf{g}_j, \mathbf{r}_j \in \mathbb{R}^N$ and decay rates $\lambda_j, e_j(\bar{\mathbf{u}}) > 0$, that need to be identified from data.

Define the hidden states as,

$$\beta_j(t) = \int_0^t \mathcal{K}_j(s,\bar{\mathbf{u}})\phi_j(s,\bar{\mathbf{u}})ds \tag{32}$$

and differentiate with respect to time,,

$$\dot{\beta}_j(t) = -\left[\lambda_j + e_j(\bar{\mathbf{u}})\right]\beta_j(t) + \mathbf{q}_j^\top \bar{\mathbf{u}}(t) + \mathbf{g}_j(\bar{\mathbf{u}})^\top \bar{\mathbf{u}}(t) + \mathbf{r}_j^\top \bar{\mathbf{f}}(t) \tag{33}$$

to obtain the memory,

$$\int_0^t \boldsymbol{\kappa}(t,s,\bar{\mathbf{u}})ds = \sum_{j=1}^{m} \left[\mathbf{p}_j + \mathbf{d}_j(\bar{\mathbf{u}})\right]\beta_j(t) \tag{34}$$

Then, Eq. (29b) is recast as the extended Markovian system,

$$\begin{aligned} \bar{\mathbf{A}}(\mathbf{w})\dot{\bar{\mathbf{u}}} &= \bar{\mathbf{B}}(\mathbf{w})\bar{\mathbf{u}} + [\mathbf{P} + \mathbf{D}(\bar{\mathbf{u}})]\,\boldsymbol{\beta} + \bar{\mathbf{f}}(t) \\ \dot{\boldsymbol{\beta}} &= [\mathbf{Q} + \mathbf{G}(\bar{\mathbf{u}})]\,\bar{\mathbf{u}} + [\mathbf{E}(\bar{\mathbf{u}}) - \boldsymbol{\Lambda}]\,\boldsymbol{\beta} + \mathbf{R}\bar{\mathbf{f}}(t) \end{aligned} \tag{35a}$$

where the data-driven operators associated to the hidden dynamics are collected as,

$$\boldsymbol{\Lambda} = \mathrm{diag}\left[\lambda_1, \lambda_2, \ldots, \lambda_m\right] \in \mathbb{R}^{m\times m}, \qquad \mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_m] \in \mathbb{R}^{N\times m} \tag{36a}$$

$$\mathbf{D}(\bar{\mathbf{u}}) = [\mathbf{d}_1(\bar{\mathbf{u}}), \mathbf{d}_2(\bar{\mathbf{u}}), \ldots, \mathbf{d}_m(\bar{\mathbf{u}})] \in \mathbb{R}^{N\times m}, \qquad \mathbf{Q} = [\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_m] \in \mathbb{R}^{m\times N} \tag{36b}$$

$$\mathbf{G}(\bar{\mathbf{u}}) = [\mathbf{g}_1(\bar{\mathbf{u}}), \mathbf{g}_2(\bar{\mathbf{u}}), \ldots, \mathbf{g}_m(\bar{\mathbf{u}})] \in \mathbb{R}^{m\times N}, \qquad \mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m] \in \mathbb{R}^{m\times N} \tag{36c}$$

$$\mathbf{E}(\bar{\mathbf{u}}) = \mathrm{diag}\left[e_1(\bar{\mathbf{u}}), e_2(\bar{\mathbf{u}}), \ldots, e_m(\bar{\mathbf{u}})\right] \in \mathbb{R}^{m\times m} \tag{36d}$$

The form of the temperature-dependent matrices $\mathbf{D}(\bar{\mathbf{u}})$, $\mathbf{G}(\bar{\mathbf{u}})$, and $\mathbf{E}(\bar{\mathbf{u}})$ is specified in the next section. Since the hidden states $\boldsymbol{\beta}$ serve as the memory, their initial conditions are set to zero, i.e., $\boldsymbol{\beta}(t_0) = \mathbf{0}$, no memory at the beginning of the time horizon. The physics-infused model in Eq. (35a) retains the structure of the LCM and depends on the displacements $\mathbf{w}$, while the hidden states account for missing physics through corrections to the stiffness and advection matrices, as well as the forcing term.

2. Coupled Physics-Infused Model

The next step involves coupling the physics-infused model in Eq. (35a) with the SRM in Eq. (10) to define the PIROM for ablating TPS. To this end, define the surface temperature $\mathbf{z}_u \in \mathbb{R}^{\tilde{N}}$ and displacements $\mathbf{z}_w \in \mathbb{R}^{\tilde{N}}$ for $\tilde{N} \le N$ ablating components so that the observable is given by $\mathbf{z} = [\mathbf{z}_u, \mathbf{z}_w]^\top \in \mathbb{R}^{n_z}$ with $n_z = 2\tilde{N}$ as the total number of observables.

Collect the RPM and hidden states into a single state vector $\mathbf{y} = [\bar{\mathbf{u}}, \mathbf{w}, \boldsymbol{\beta}]^\top \in \mathbb{R}^{n_y}$, where $n_y = N + \tilde{N} + m$, and define a data-driven operator $\mathbf{M} \in \mathbb{R}^{n_z \times n_y}$ to define the PIROM observable as,

$$\mathbf{z} = \mathbf{M}\mathbf{y} \tag{37}$$

where,

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_{\bar{u}} & \mathbf{0} & \mathbf{M}_{\beta} \\ \mathbf{0} & \mathbf{I} & \mathbf{0} \end{bmatrix} \tag{38}$$

includes the matrices $\mathbf{M}_{\bar{u}} \in \mathbb{R}^{\tilde{N}\times N}$ and $\mathbf{M}_{\beta} \in \mathbb{R}^{\tilde{N}\times m}$, which computes the surface temperature observable from the

RPM states and hidden states, respectively. The PIROM is coupled to the SRM in Eq. (10) by leveraging Eq. (37) to compute the surface recession velocity. Thus, the PIROM is formally stated as,

$$\mathcal{A}\dot{\mathbf{y}} = [\mathcal{B} + \mathcal{C}]\,\mathbf{y} + \mathcal{F}(t) \tag{39a}$$

$$\mathbf{z} = \mathbf{M}\mathbf{y} \tag{39b}$$

where,

$$\mathcal{A} = \begin{bmatrix} \bar{\mathbf{A}}(\mathbf{w}) & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{I} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{I} \end{bmatrix} \in \mathbb{R}^{n_y \times n_y}, \quad \mathcal{B} = \begin{bmatrix} \bar{\mathbf{B}}(\mathbf{w}) & \mathbf{O} & \mathbf{P} \\ \mathbf{\Xi}\mathbf{M}_u & \mathbf{O} & \mathbf{\Xi}\mathbf{M}_\beta \\ \mathbf{Q} & \mathbf{O} & -\mathbf{\Lambda} \end{bmatrix} \in \mathbb{R}^{n_y \times n_y}, \tag{40a}$$

$$\mathcal{C} = \begin{bmatrix} \mathbf{O} & \mathbf{O} & \mathbf{D}(\bar{\mathbf{u}}) \\ \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{G}(\bar{\mathbf{u}}) & \mathbf{O} & \mathbf{E}(\bar{\mathbf{u}}) \end{bmatrix} \in \mathbb{R}^{n_y \times n_y}, \quad \mathcal{F} = \begin{bmatrix} \bar{\mathbf{f}}(t) \\ -\tilde{\mathbf{f}} \\ \mathbf{R}\bar{\mathbf{f}}(t) \end{bmatrix} \in \mathbb{R}^{n_y} \tag{40b}$$

The learnable parameters in the PIROM are collected as,

$$\mathbf{\Theta} = \{\mathbf{P}, \mathbf{\Lambda}, \mathbf{Q}, \mathbf{M}, \mathbf{D}(\bar{\mathbf{u}}), \mathbf{G}(\bar{\mathbf{u}}), \mathbf{E}(\bar{\mathbf{u}}), \mathbf{R}\}\,, \in \mathbb{R}^{n_\theta} \tag{41}$$

The matrices $\mathbf{P}, \mathbf{\Lambda}, \mathbf{Q}, \mathbf{M}, \mathbf{R}$ are constants, and account for the effects of coarse-graining on the stiffness, output, and forcing matrices. The matrices $\mathbf{D}(\bar{\mathbf{u}}), \mathbf{E}(\bar{\mathbf{u}}), \mathbf{G}(\bar{\mathbf{u}})$ are temperature-dependent matrices, and account for the effects of coarse-graining on the advection matrix due to mesh motion. Leveraging the DG-FEM formula for the advection matrix in Eq. (56c) in Appendix A.A, and noting that the ablating velocity in Eq. (4) imposes the boundary condition for the mesh motion, the state-dependent matrices for the i-th component are written as,

$$\mathbf{D}(\bar{\mathbf{u}}) \approx \mathrm{diag}\,[\dot{\mathbf{w}}]\,\mathbf{D}, \quad \mathbf{G}(\bar{\mathbf{u}}) \approx \mathbf{G}\mathrm{diag}\,[\dot{\mathbf{w}}]\,, \quad \mathbf{E}(\bar{\mathbf{u}}) \approx \mathrm{diag}\begin{bmatrix} \underbrace{\dot{\mathbf{w}}, \dots, \dot{\mathbf{w}}}_{\tilde{m} \text{ times}} \end{bmatrix} \mathbf{E} \tag{42}$$

where $\dot{\mathbf{w}} = \dot{\mathbf{w}}(\bar{\mathbf{u}})$ is the SRM based on the observable temperature $\bar{\mathbf{u}}$ and $\tilde{m}$ is the number of hidden states per component so that $m = N\tilde{m}$.

The PIROM in Eq. (39b) incorporates explicit information on the material properties, boundary conditions, and surface recession, and is designed to generalize across parametric variations in these inputs. Moreover, the hidden dynamics in Eq. (35a) are interpretable, as these retain the functional form of the DG-FEM in Eq. (7). The next step is focused on identifying the unknown data-driven parameters $\mathbf{\Theta}$ characterizing the hidden dynamics.

C. Learning the Hidden Dynamics

Learning of the PIROM is achieved through a gradient-based neural-ODE-like (NODE) approach [8]. For ease of presentation, consider the compact form of the PIROM in Eq. (39b),

$$\mathcal{D}\left(\dot{\mathbf{y}}, \mathbf{y}, \boldsymbol{\xi}, \mathcal{F}; \mathbf{\Theta}\right) = \mathbf{0} \tag{43}$$

where $\boldsymbol{\xi}$ defines the model parameters, i.e., SRM parameters, while $\mathcal{F}$ represents the forcing terms, i.e., the boundary conditions due to heating.

Consider a dataset of N_s high-fidelity *surface temperature* observable trajectories $\mathbf{z}_{\mathrm{HF}}$, sampled at p time instances $\{t_k\}_{k=0}^{p-1}$, for different parameter settings $\{\boldsymbol{\xi}^{(l)}\}_{l=1}^{N_s}$ and forcing functions $\{\mathcal{F}^{(l)}(t)\}_{l=1}^{N_s}$. The dataset is expressed as,

$$\mathcal{D} = \left\{\left(t_k, \mathbf{z}_{\mathrm{HF}}^{(l)}(t_k), \boldsymbol{\xi}^{(l)}, \mathcal{F}^{(l)}(t_k)\right)\right\}_{l=1}^{N_s}, \quad k = 0, 1, \dots, p-1 \tag{44}$$

In this work, the dataset contains only surface temperature observables – all high-fidelity information regarding the surface displacements *are assumed to be unavailable during learning*.

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The learning problem is formulated as the following differentially-constrained problem,

$$\min_{\mathbf{\Theta}} \quad \mathcal{J}(\mathbf{\Theta}; \mathcal{D}) = \sum_{l=1}^{N_s} \int_{t_0}^{t_f} \ell\left(\mathbf{z}_u^{(l)}, \mathbf{z}_{\mathrm{HF}}^{(l)}\right) dt \tag{45a}$$

$$\text{s.t.} \quad \mathbf{0} = \mathcal{D}\left(\dot{\mathbf{y}}^{(l)}, \mathbf{y}^{(l)}, \boldsymbol{\xi}^{(l)}, \mathcal{F}^{(l)}; \mathbf{\Theta}\right) \tag{45b}$$

for $l = 1, 2, \ldots, N_s$, the objective is to minimize the discrepancy between the high-fidelity and PIROM predictions for the l-th trajectory with $\ell\left(\mathbf{z}_u^{(l)}, \mathbf{z}_{\mathrm{HF}}^{(l)}\right) = \left\|\mathbf{z}_u^{(l)} - \mathbf{z}_{\mathrm{HF}}^{(l)}\right\|_2^2$.

The gradient-based optimization loop is based on the adjoint variable $\boldsymbol{\lambda}$, governed by the adjoint differential equation,

$$\begin{aligned} \frac{\partial \ell}{\partial \mathbf{y}} + \boldsymbol{\lambda}^\top \frac{\partial \mathcal{D}}{\partial \mathbf{y}} - \frac{d}{dt}\left(\boldsymbol{\lambda}^\top \frac{\partial \mathcal{D}}{\partial \dot{\mathbf{y}}}\right) &= \mathbf{0} \\ \boldsymbol{\lambda}(t_f)^\top \frac{\partial \mathcal{D}}{\partial \dot{\mathbf{y}}(t_f)} &= \mathbf{0} \end{aligned} \tag{46a}$$

Once $\boldsymbol{\lambda}$ is solved, the gradient is computed as,

$$\nabla_{\mathbf{\Theta}} \mathcal{J} = \frac{1}{N_s} \sum_{l=1}^{N_s} \int_{t_0}^{t_f} \left(\frac{\partial \ell}{\partial \mathbf{\Theta}} + \left(\boldsymbol{\lambda}^{(l)}\right)^\top \frac{\partial \mathcal{D}}{\partial \mathbf{\Theta}}\right) dt \tag{47}$$

The PIROM parameters $\mathbf{\Theta}$ are updated via stochastic gradient descent using ML optimizers such as Adam. The learning procedure iterates between solving the PIROM in Eq. (43) forward in time, solving the adjoint equation in Eq. (46a) backward in time, and updating the parameters $\mathbf{\Theta}$ until convergence.

IV. Application to Thermal Protection Systems

In this section, the proposed PIROM approach is applied to the analysis of thermo-ablative multi-layered hypersonic TPS. The performance of the PIROM is evaluated in terms of the three corners of the impossible trinity of modeling in Fig. 1, based on parametric variations of boundary conditions and SRMs. The results show PIROM to be a promising candidate for solving the impossible trinity of modeling, achieving RPM-level computational efficiency and generalizability with FOM-level accuracy.

A. Problem Definition

Consider the two-dimensional TPS configuration shown in Fig. 3 with constant material properties within each layer, dimensions, and BCs listed in Table 1. This configuration is representative of the TPS typically used for high-speed vehicles, and involves two main layers: an outer ablative layer, and an inner substrate layer. The ablative layer, composed of $\tilde{N} = 3$ ablative components, is subjected to strong time-varying and non-uniform heating, while the substrate layer, composed of one non-ablative component, is insulated adiabatically at the outer surface; the total number of components is thus $N = 4$.

The lumped-mass representation of the TPS configuration is shown in Fig. 3(b), where each component Ω_i is represented by a lumped mass with uniform temperature $\bar{u}_i(t)$. Details regarding the derivation of the LCM for this configuration are provided in Appendix A.C. The sources of non-linearities studied in this problem originate from the coupling between the thermodynamics and temperature-dependent mesh motion, i.e., geometry-dependent matrices and temperature advection, as well as the heterogeneities across material layers. As shown in Fig. 3, perfect thermocouple devices are placed at the surfaces of the ablative layers for the collection of the high-fidelity temperature signals that are used in the following sections for training and testing the PIROM.

B. Problem Parametrization

The operating conditions of the TPS are specified by the boundary conditions, i.e., the heat flux, and the SRM matrix $\mathbf{\Xi}$. Specifically, the heat flux on the Neumann BC is parametrized using $\boldsymbol{\xi}_{\mathrm{BC}} = \{\xi_1, \xi_2, \xi_3\}$, while the SRM is parametrized using $\boldsymbol{\xi}_{\mathrm{SRM}} = \{\alpha_1, \alpha_2, \alpha_3\}$. Thus, the heat flux and SRM over the i-th ablative component are expressed

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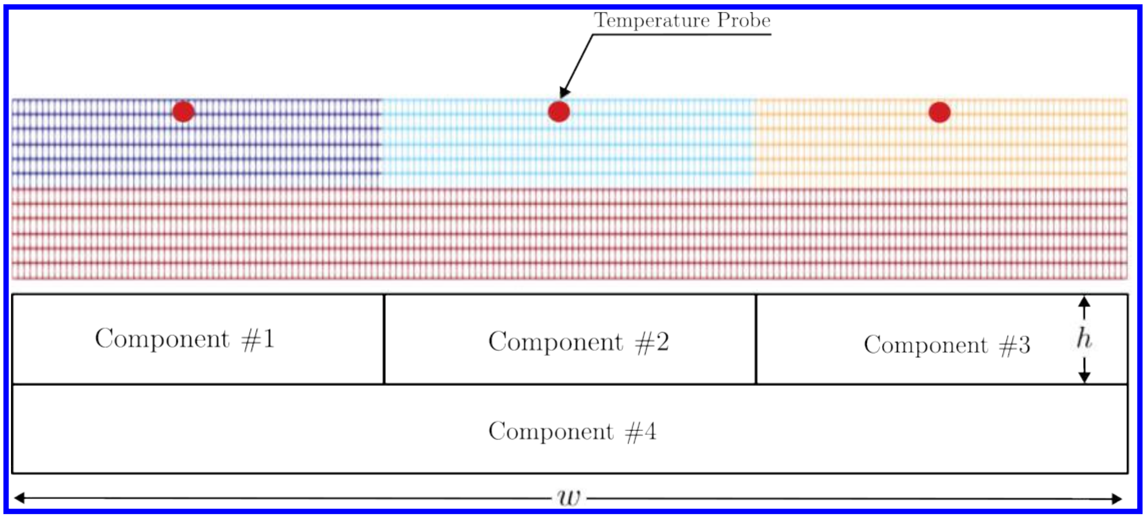
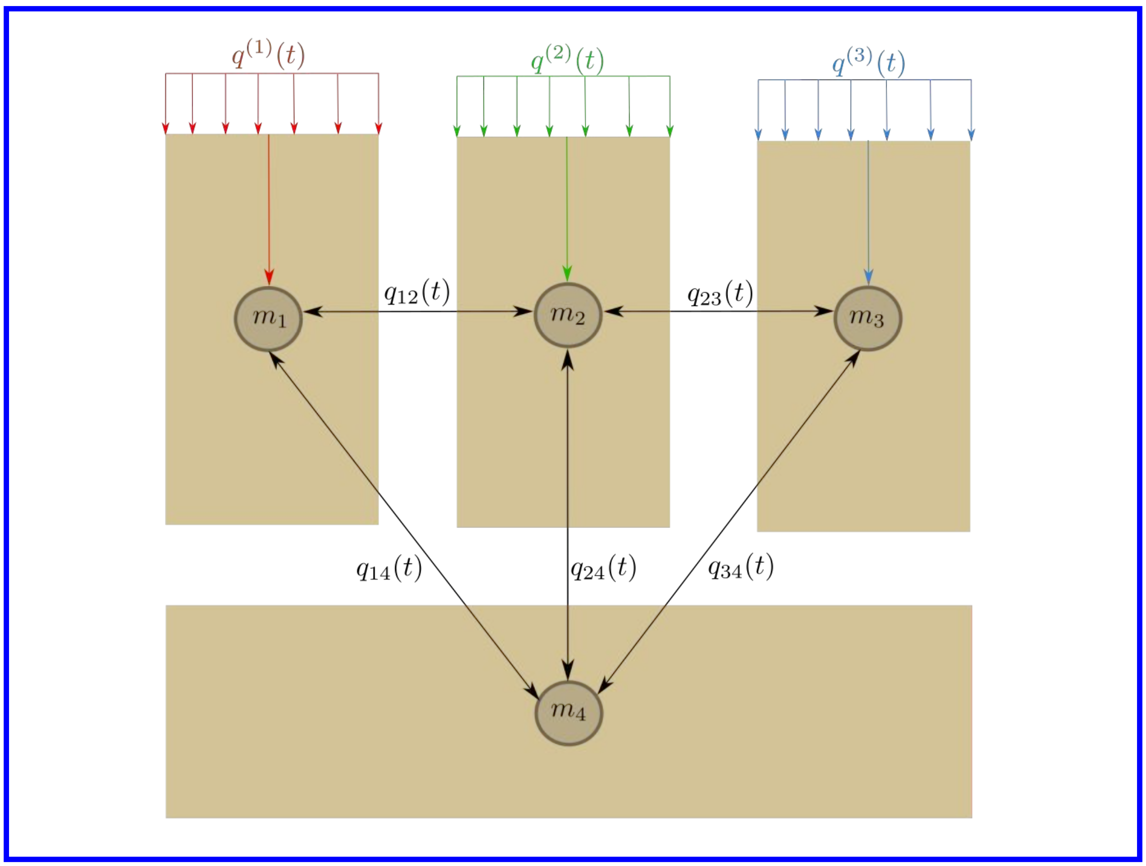


(a) Mesh and Geometry



(b) Lumped Mass Representation

Fig. 3 Four-component TPS geometry and lumped-mass representation for the TPS.

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Component	w (cm)	h (cm)	ρ (kg/m³)	c_p (J/kg·K)	k (W/m·K)	$\alpha \times 10^{-6}$ (m/s·K)
#1	0.3	0.03	160	1200	0.2	1
#2	0.3	0.03	1800	900	5	1
#3	0.3	0.03	300	1500	0.15	1
#4	0.9	0.03	1600	800	10	0

Table 1 Description of TPS components, including thickness h, density ρ, specific heat capacity c_p, thermal conductivity k, and SRM parameter α.

	Parameters					
Dataset	$\xi_1 \times 10^3$	$\xi_2 \times 10^{-1}$	$\xi_3 \times 10^{-2}$	$\alpha_1 \times 10^{-6}$	$\alpha_2 \times 10^{-6}$	$\alpha_3 \times 10^{-6}$
$\mathcal{D}_1$	$[-6.059, -5.902]$	$[-3.501, 3.152]$	$[9.670, 10.464]$	1	1	1
$\mathcal{D}_2$	$[-6.122, -5.887]$	$[-3.601, -3.074]$	$[9.218, 11.246]$	1	1	1
$\mathcal{D}_3$	6	-3.333	10	$[0.6, 1.5]$	$[0.6, 1.5]$	$[0.6, 1.5]$

Table 2 Range of parameters $[\min, \max]$ **in training and testing datasets.**

as,

$$q_i(x, t; \boldsymbol{\xi}_{\mathrm{BC}}) = \xi_1 e^{\xi_2 x} e^{\xi_3 t},\ \forall x \in \Gamma_{i,q}, \quad \dot{w}_i(z_{u,i}; \boldsymbol{\xi}_{\mathrm{SRM}}) = \alpha_i \left(z_{u,i} - u_{0,i}\right),\ i = 1, \ldots, \tilde{N} \tag{48}$$

where $\Gamma_{i,q}$, $z_{u,i}$, and $u_{0,i}$ correspond to the Neumann BC surface, the surface temperature prediction, and the initial temperature of the i-th ablative component, respectively. The parameters ξ_1, ξ_2, and ξ_3 control the heat flux magnitude, spatial variation, and temporal variation, respectively. The constant α_i is a small material-dependent constant determined from the B' table [17], specifying the surface recession velocity for a given temperature.

C. Data Generation

Full-order solutions of the TPS are computed using the FEM multi-mechanics module from SIERRA/Aria with the mesh shown in Fig. 3(a) [14]. The mesh consists of 2196 total elements, with 366 elements for each ablative component and 1098 elements for the substrate component. Given an operating condition $\boldsymbol{\xi} = \left[\boldsymbol{\xi}_{\mathrm{BC}}, \boldsymbol{\xi}_{\mathrm{SRM}}\right]^\top$, a high-fidelity solution is computed for one minute, starting from an uniform initial temperature of $T(x, t_0) = 300$ K. Each solution consist of a collection of space-time-varying temperature and displacement fields $\left\{\left(t_k, \mathbf{u}_{\mathrm{HF}}^{(l)}(t_k), \mathbf{d}_{\mathrm{HF}}^{(l)}(t_k), \boldsymbol{\xi}^{(l)}\right)\right\}_{k=0}^{p-1}$, where p is the number of time steps with a step size of $\Delta t \approx 10^{-3}$. The observable trajectories are representative of near-wall thermocouple sensing of hypersonic flows involving heat transfer. At each time instance t_k, a temperature reading is recorded from each ablative component using the thermocouples shown in Fig. 3, resulting in three temperature signals, i.e., the observables $\mathbf{z}_{\mathrm{HF}} \in \mathbb{R}^3$. Therefore, each full-order solution produces one trajectory of observables $\left\{\left(t_k, \mathbf{z}_{\mathrm{HF}}^{(l)}(t_k), \boldsymbol{\xi}^{(l)}\right)\right\}_{k=0}^{p-1}$. The goal of the PIROM is to predict the surface temperature and displacement as accurately as possible.

1. Definition of Training and Testing Datasets

The range of parameters used to generate the training $\mathcal{D}_1$ and testing $\{\mathcal{D}_2, \mathcal{D}_3\}$ datasets are listed in Table 2. The training and testing datasets are designed, respectively, to: (1) minimize the information that the PIROM can "see", and (2) to maximize the variability of test operating conditions to examine the PIROM's generalization performance. A total of 110 normally-distributed data points for the BC parametrization are visualized in Fig. 4(a), and the corresponding observable trajectories are shown in Figs. 4(b) and 4(c). The training dataset $\mathcal{D}_1$ includes 10 trajectories with randomly selected BC parameters from the 110 points, with nominal SRM parameters $\boldsymbol{\xi}_{\mathrm{SRM}} = \{1, 1, 1\} \times 10^{-6}$. Note that although Fig. 4(c) shows the surface displacements for all ablative components in $\mathcal{D}_1$, only the *surface temperature is used for training the PIROM*.

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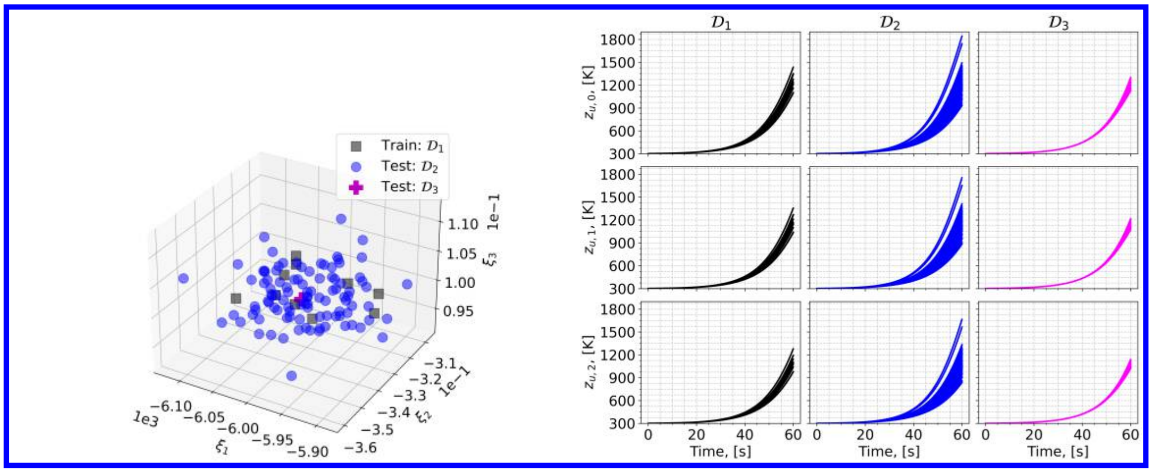
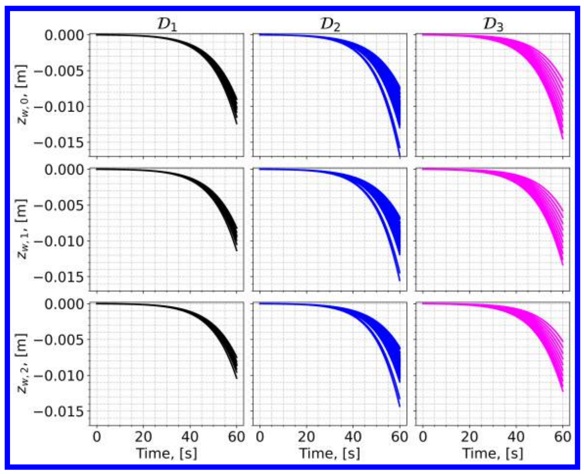


(a) BC Parameters (b) Surface Temperatures



(c) Surface Displacements

Fig. 4 Boundary condition parameters, temperature observables, and displacement observables for the training and testing datasets. The variables $z_{u,i}$ and $z_{w,i}$ correspond to the surface temperature and displacement of the i-th ablative component, respectively.

Two additional datasets are generated for testing. The dataset $\mathcal{D}_2$ includes the remaining 100 BC parameter values not considered in $\mathcal{D}_1$, with the same nominal SRM parameters. The cases in the $\mathcal{D}_3$ fixes the boundary condition as shown in Fig. 4(a) and varies the SRM parameters as shown in Table. 2. The testing dataset $\mathcal{D}_3$ includes *out-of-distribution* (OOD) trajectories, and is designed to test the generalizability of the PIROM to unseen SRMs.

D. Performance Metrics

The performance metrics are defined to quantitatively assess the solution to the impossible trinity of modeling for the TPS problem. Specifically, the *accuracy* metric quantifies the prediction error of the ROMs against high-fidelity solutions. The *efficiency* metric quantifies the computational speedup achieved by the ROMs compared to high-fidelity simulations. The *generalizability* metric quantifies the ability of the ROMs to retain accuracy when evaluated on OOD datasets. Together, these metrics provide a comprehensive evaluation of the PIROM's performance in addressing the challenges associated with modeling complex multi-physics systems. Since the generalizability metric is inherently tied to the accuracy metric when evaluated on OOD datasets, the following sections focus on defining the accuracy and efficiency metrics.

Accuracy Metric Consider one trajectory of high-fidelity surface temperature and displacement data,

$$\left\{\left(t_k, \mathbf{z}_{u,\mathrm{HF}}^{(l)}(t_k), \mathbf{z}_{w,\mathrm{HF}}^{(l)}(t_k)\right)\right\}_{k=0}^{p-1}$$

for the l-th operating condition in the testing datasets $\mathcal{D}_2$ or $\mathcal{D}_3$. The difference $e_i^{(l)}$ for the i-th predicted observable, denoted as $z_i^{(l)}$, is computed as,

$$e_i^{(l)} = \frac{1}{\Delta z_i^{(l)}} \sqrt{\frac{1}{p} \sum_{k=0}^{p-1} \left(z_{i,\mathrm{HF}}^{(l)}(t_k) - z_i^{(l)}(t_k)\right)^2} \tag{49}$$

for $i = 1, 2, 3$ and $z_i^{(l)} \in \left\{z_{i,u}^{(l)}, z_{i,w}^{(l)}\right\}$, and where,

$$\Delta z_i^{(l)} = \max_{0 \le k \le p-1} z_{i,\mathrm{HF}}^{(l)}(t_k) - \min_{0 \le k \le p-1} z_{i,\mathrm{HF}}^{(l)}(t_k) \tag{50}$$

Subsequently, the prediction error of one trajectory is computed by a weighted sum based on the area of each *ablative component*, resulting in the normalized root mean square error (NRMSE) metric for one trajectory,

$$\mathrm{NRMSE} = \frac{\sum_{i=1}^{\tilde{N}} |\Omega_i|\, e_i^{(l)}}{\sum_{i=1}^{\tilde{N}} |\Omega_i|} \times 100\% \tag{51}$$

For one dataset, the NRMSE is defined to be the average of the NRMSEs of all trajectories in the dataset.

Efficiency Metric The efficiency metric is quantified using the *computational acceleration*, which focuses on the quantification of the spedup factor $\frac{\mathcal{T}_{\mathrm{HF}}(\mathcal{D})}{\mathcal{T}_{\mathcal{M}}(D)}$. The terms $\mathcal{T}_{\mathrm{HF}}(\mathcal{D})$ and $\mathcal{T}_{\mathcal{M}}(\mathcal{D})$ correspond to the wall-clock time required by the high-fidelity model and the ROM, to evaluate all trajectories in the dataset $\mathcal{D}$, respectively. Here, $\mathcal{M}$ corresponds to the ROM under consideration, i.e., either the PIROM or the RPM. On the training phase, our previous work has shown that the PIROM training cost is comparable to that of training a NODE under equivalent training datasets and optimization algorithms [10].

E. Generalization to Boundary Conditions

To assess generalization to BC, the PIROM and RPM are evaluated on the $\mathcal{D}_2$ dataset. Temperature trajectory predictions for a representative test case are shown in Figs. 5(a) and 5(b), where the PIROM accurately captures the surface temperature and displacement dynamics, while the RPM exhibits larger deviations and under-predicts surface displacements due to the averaging effects of the LCM. The mean NRMSE across all test cases in $\mathcal{D}_2$ is shown in Figs. 5(e) and 5(f), where the PIROM consistently achieves errors of 1% for both temperature and displacement predictions, improving the RPM's accuracy by an order of magnitude. Figure 5 reports the average substrate temperature, where the LCM remains highly accurate due to the symmetric TPS geometry, adiabatic BCs, and negligible thermal gradients within the substrate. Although the PIROM is trained only on the surface temperatures of the three ablative components, its hidden dynamics retain the LCM's accuracy for this untrained observable, demonstrating the PIROM's ability to generalize and preserve the underlying physics of the reduced-physics backbone. The consistently low prediction errors demonstrate the solution to the *accuracy* corner of the impossible trinity of modeling.

F. Generalization to Surface Recession Models

The generalization performance of the PIROM and RPM is also evaluated on surface recession models using the OOD $\mathcal{D}_3$ dataset. As detailed in Table 1, the SRM parameter α in $\mathcal{D}_3$ is perturbed 10 times by up to ±50% from their nominal values. The SRM model perturbation introduces significant changes to the ablative layer dynamics, potentially increasing the rate of ablation at lower temperatures, as shown in Figs. 5(c) and 5(d). The PIROM, without considering any SRM variations during training, is able to accurately predict the surface temperature and displacement dynamics for the perturbed SRMs. Figures 5(e) and 5(f) show the mean NRMSE across all test cases in $\mathcal{D}_3$, where the PIROM consistently achieves errors below 1% for both temperature and displacement predictions, and consistently improves the RPM's accuracy by approximately an order of magnitude. The consistent low prediction errors demonstrate the solution

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to the *generalizability* corner of the impossible trinity of modeling.

G. Computational Cost

All computations are performed in serial for fairness on an `Intel Xeon (R) Gold 6258R CPU 2.70GHz` computer with 62 GB of RAM. The numerical integration for the RPM and PIROM models are performed using `SciPy`'s `solve_ivp` function with default settings. Provided a parametrization for the BC and SRM, the high-fidelity FEM simulation takes about ≈ 60 seconds, the RPM takes about ≈ 0.137 seconds, and the PIROM takes about ≈ 0.280 seconds. Therefore, during evaluation both the RPM and PIROM achieve speedup factors of approximately 438 and 214, respectively, over the high-fidelity model. As a result, the PIROM and RPM are *two-orders-of-magnitude faster* than the high-fidelity model. The PIROM nearly preserves the computational efficiency of the RPM (about twice as expensive as the RPM), while achieving significantly higher accuracy and generalization capabilities. The results demonstrate the benefits of physics-infused modeling for the development of efficient and generalizable ROMs for complex multi-physics systems, and demonstrate the solution to the *efficiency* corner of the impossible trinity of modeling.

H. Summary of Results

The results presented in this section demonstrate the accuracy, generalizability, and computational efficiency of the proposed PIROM approach for the analysis of thermo-ablative multi-layered hypersonic TPS. The PIROM consistently achieves low prediction errors below 1% for both surface temperature and displacement across a range of unseen boundary conditions and surface recession models. Furthermore, the PIROM retains the computational efficiency of traditional RPMs, achieving speedup factors of over 200 times compared to high-fidelity FEM simulations. The generalization capabilities of the PIROM are attributed to its hybrid structure: a physics-based LCM backbone that ensures consistency with the underlying thermodynamics, while a data-driven correction mechanism captures the un-resolved dynamics. For this TPS problem, the PIROM successfully addresses the impossible trinity of modeling, achieving high-fidelity model accuracy, RPM-level computational efficiency, and generalizability to unseen operating conditions.

V. Conclusions

This work presents the development and validation of the *scientific machine learning* framework termed *Physics-Informed Reduced Order Model* (PIROM) for simulating the transient thermo-ablative response of hypersonic thermal protection systems (TPS) subjected to hypersonic flow. Using coarse-graining on a DG-FEM model and the Mori-Zwanzig formalism, the PIROM formulation in Ref. [10] is extended to account for non-decomposing thermo-ablative response of a multi-layered TPS. The PIROM builds upon the following two key components: (1) a first-order physics-based model, i.e., the RPM based on LCM and SRM, for low-fidelity predictions of the surface temperature and recession; and (2) a data-driven closure to the non-Markovian term in the Generalized Langevin Equation (GLE). The non-Markovian closure is recast as a set of hidden states that evolve according to a data-driven dynamical system that is learned from a sparse collection of high-fidelity temperature signals.

The results demonstrate that the PIROM framework effectively reconciles the trade-offs between accuracy, generalizability, and efficiency of the ITM for simulating ablating hypersonic TPS. The PIROM consistently achieves mean observable prediction errors of less than 1% for extrapolative settings involving time-and-space varying boundary conditions and SRM models. Notably, the PIROM improves the RPM's accuracy by an order of magnitude while preserving its computational efficiency, physical interpretability, and parametric generalizability. Moreover, the PIROM delivers online evaluations that are two orders of magnitude faster than the FOM. These results highlight the PIROM's potential as a promising framework for optimizing multi-physical dynamical systems, such as TPS under diverse operating conditions.

Future work will focus on reducing the training costs, which currently depend on trajectory optimization and continuous adjoint-based gradient evaluations. To mitigate this computational buren, strategies such as large-scale parallelization and weak-form learning [18] offer promising avenues for acceleration. In addition, the development of rigorous theoretical convergence guarantees remains an open area of investigateion and is essential for establishing broader confidence in the PIROM's reliability and scalability.

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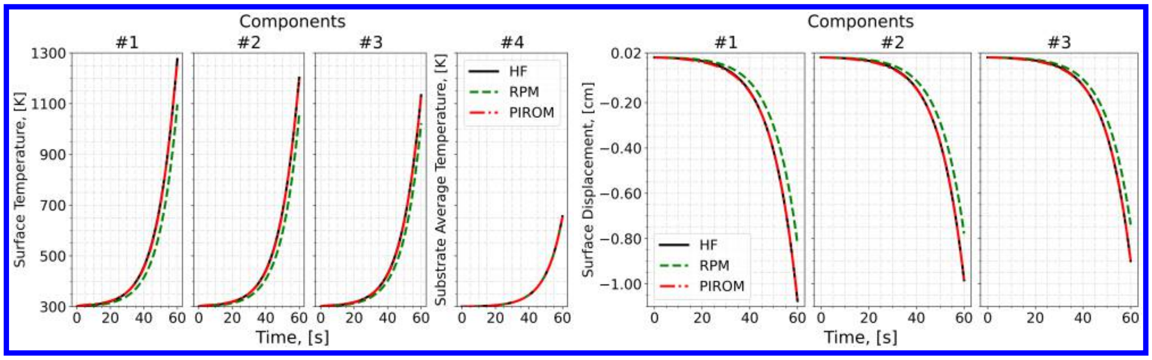
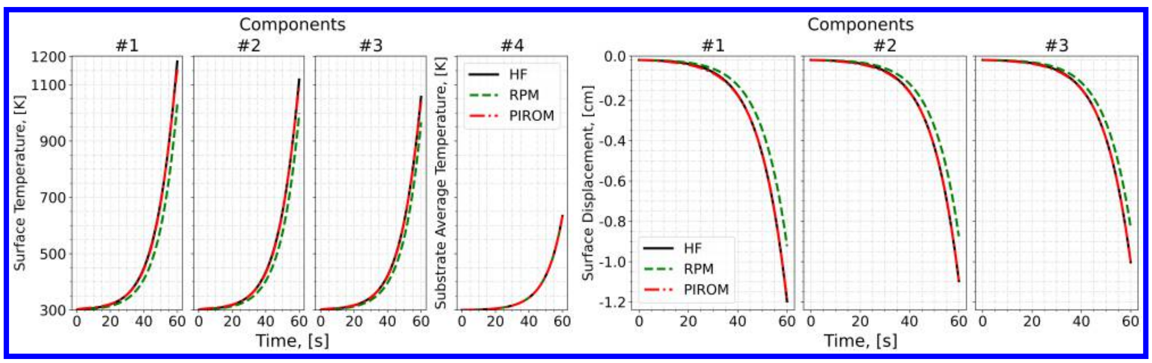


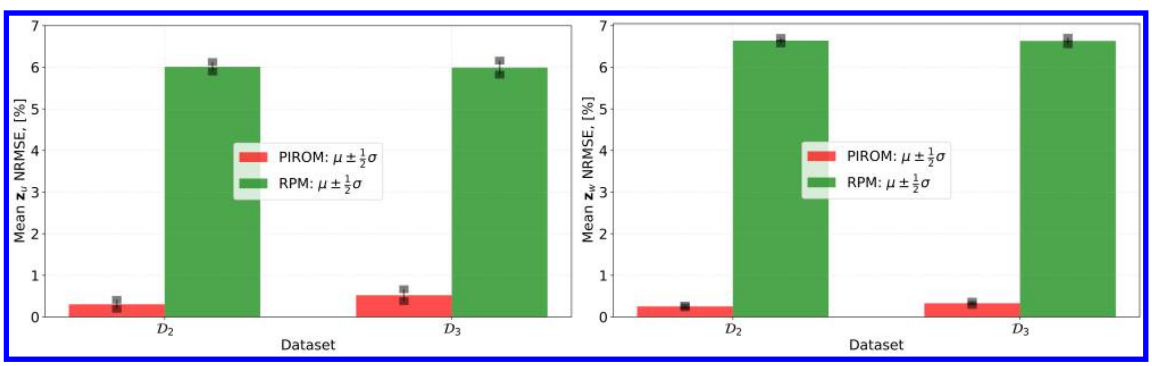
(a) BC Generalization

(b) BC Generalization



(c) SRM Generalization

(d) SRM Generalization



(e) Mean Temperature Errors

(f) Mean Displacement Errors

Fig. 5 PIROM predictions against high-fidelity solutions for (a)-(b) BC generalization, (c)-(d) SRM generalization, and (e)-(f) mean errors across testing datasets.

A. Technical Details

This appendix presents the technical details of the PIROM framework applied to the transient modeling of thermo-ablative TPS. The first section provides the mathematical details for the definition of the DG-FEM. The second section details the coarse-graining procedures performed on the DG-FEM representation of the TPS. The third section presents the derivation of the LCM model from an energy-conservation perspective.

A. Full-Order Model

To obtain the full-order numerical solution, the governing equation is spatially discretized using variational principles of Discontinuous Galerkin (DG) to result in a high-dimensional system of ordinary differential equations (ODEs). The DG-FEM model is written in an element-wise form, which is beneficial for subsequent derivations of the lower-order models. Note that the choice of DG approach here is mainly for theoretical convenience in the subsequent coarse-graining formulation. In the numerical results, the full-order TPS ablation simulations is computed using standard FEM instead, and the equivalence between DG and standard FEM is noted upon their convergence.

1. Domain Discretization

Consider a conforming mesh partition of the domain in Fig. 3, where each element belongs to one and only one component. Denote the collection of all M elements as $\{E_i\}_{i=1}^{M}$. To ease the description of the DG model, a graph structure is employed. The elements are treated as vertices, the set of which is denoted $\mathcal{V} = \{m\}_{m=1}^{M}$. Two neighboring elements, E_i and E_j, are connected by an edge (i, j), and the shared boundary between them is denoted e_{ij}. The collection of all edges are denoted $\mathcal{E}$, and $\mathcal{G}$ is referred to as a graph. In the graph, the edges are unidirected, meaning if $(i, j) \in \mathcal{E}$ then $(j, i) \in \mathcal{E}$. Furthermore, denote the neighbors of the i-th element as $\mathcal{N}_i = \{j|(i, j) \in \mathcal{E}\}$. Lastly, for the ease of notation, introduce two special indices: T for the boundary of an element that overlaps with the Dirichlet boundary condition, and similarly q for the Neumann boundary condition.

2. Weak Form of Discontinuous Galerkin Method

Choosing appropriate basis functions ϕ_k and ϕ_l and using the Interior Penalty Galerkin (IPG) scheme [3], the variational bilinear form for Eq. (1a) is,

$$\sum_{i=1}^{M} a_{\epsilon,i}(\phi_k, \phi_l) = \sum_{i=1}^{M} L_i(\phi_k) \tag{52}$$

where ϵ is an user-specified parameter and,

$$a_{\epsilon,i}(\phi_k, \phi_l) = \int_{E_i} \left(\rho c_p \phi_k \frac{\partial \phi_l}{\partial t} + \mathbf{k}\nabla\phi_k \cdot \nabla\phi_l - \rho c_p \phi_k \mathbf{v} \cdot \nabla\phi_l\right) dE_i \tag{53a}$$

$$- \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \int_{e_{ij}} \{\mathbf{k}\nabla\phi_k \cdot n\} [\phi_l]\, de_{ij} + \epsilon \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \int_{e_{ij}} \{\mathbf{k}\nabla\phi_l \cdot n\} [\phi_k]\, de_{ij}$$

$$+ \sigma \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \int_{e_{ij}} [\phi_k] [\phi_l]\, de_{ij} \tag{53b}$$

$$L_i(v) = \epsilon \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \int_{e_{ij}} (\mathbf{k}\nabla\phi_l \cdot n)\, T_b de_{ij} + \int_{e_{iq}} \phi_k q_b de_{iq} + \sigma \int_{e_{iT}} \phi_k T_b de_{iT} \tag{53c}$$

In the bi-linear form above, the notations [] and {} are respectively the jumps and averages at the boundary e_{ij} share by two elements E_i and E_j,

$$[u] = u\big|_{E_i} - u\big|_{E_j}, \quad \{u\} = \frac{1}{2}\left(u\big|_{E_i} + u\big|_{E_j}\right), \quad \text{for } x \in e_{ij} = E_i \cap E_j$$

Moreover, the terms associated with σ are introduced to enforce the Dirichlet boundary conditions; σ is a penalty factor whose value can depend on the size of an element. Depending on the choice of ϵ, the bi-linear form corresponds to symmetric IPG ($\epsilon = -1$), non-symmetric IPG ($\epsilon = 1$), and incomplete IPG ($\epsilon = 0$). All these schemes are consistent

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with the original PDE and have similar convergence rate with respect to mesh size. In the following derivations, the case $\epsilon = 0$ is chosen for the sake of simplicity.

3. Discontinuous Galerkin Model

Next, the DG-FEM is written in an element-wise form. For the i-th element, use a set of P trial functions to approximate the temperature as in Eq. (6). Without loss of generality, the trial functions are assumed to be orthogonal, so that,

$$\int_{E_i} \phi_l^i(x)\phi_k^i(x)d\Omega = 0, \quad l \neq k$$

where $|E_i|$ is the area ($n_d = 2$) or volume ($n_d = 3$) of the i-th element, and δ_{lk} is the Kronecker delta function. Furthermore, for simplicity, choose $\phi_1^i = 1$. Thus, by orthogonality,

$$\int_{E_i} \phi_1^i(x)d\Omega = |E_i|, \quad \int_{E_i} \phi_k^i(x)d\Omega = 0, \quad k = 2, 3, \ldots, P$$

Under the choice of basis functions, u_1^i is simply the average temperature of element E_i, denoted as $\bar{u}_i$.

Using test functions same as trial functions, the dynamics $\mathbf{u}_i$ is obtained by evaluating the element-wise bi-linear forms,

$$a_{\epsilon,i}(\phi_k^i, T^i) = L_i(\phi_k^i), \quad k = 1, 2, \ldots, P \tag{54}$$

Therefore, by standard variational principles, e.g., [3], the element-wise governing equation is denoted as,

$$\mathbf{A}^i\dot{\mathbf{u}}_i = \left[\mathbf{B}^i + \mathbf{C}^i(\mathbf{u}_i)\right]\mathbf{u}_i + \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \left(\mathbf{B}_{ij}^i\mathbf{u}_i + \mathbf{B}_{ij}^j\mathbf{u}_j\right) + \mathbf{f}_i(t), \quad \text{for } i = 1, 2, \ldots, M \tag{55}$$

where for $k, l = 1, 2, \ldots, P$,

$$\left[\mathbf{A}^i\right]_{kl} = \int_{E_i} \rho c_p \phi_k^i \phi_l^i dE_i \tag{56a}$$

$$\left[\mathbf{B}^i\right]_{kl} = -\int_{E_i} \left(\nabla\phi_k^i\right)\cdot\left(\mathbf{k}\nabla\phi_l^i\right) dE_i \tag{56b}$$

$$\left[\mathbf{C}^i\right]_{kl} = \int_{E_i} \rho c_p \phi_k^i \mathbf{v}_m^i \cdot \nabla\phi_l^i dE_i \tag{56c}$$

$$\left[\mathbf{B}_{ij}^i\right] = \int_{e_{ij}} \left\{\mathbf{k}\nabla\phi_k^i\cdot\hat{n}\right\}\phi_l^i - \sigma\left[\phi_k^i\right]\phi_l^i de_{ij} \tag{56d}$$

$$\left[\mathbf{B}_{ij}^j\right] = \int_{e_{ij}} -\left\{\mathbf{k}\nabla\phi_k^i\cdot\hat{n}\right\}\phi_l^j + \sigma\left[\phi_k^i\right]\phi_l^j de_{ij} \tag{56e}$$

$$\left[\mathbf{f}^i\right]_k = \int_{e_{iq}} \phi_k^i q_b de_{iq} + \sigma\int_{e_{iT}} \phi_k^i de_{iT} \tag{56f}$$

The matrices $\mathbf{A}^i \in \mathbb{R}^{P\times P}$, $\mathbf{B}^i \in \mathbb{R}^{P\times P}$, and $\mathbf{C}^i \in \mathbb{R}^{P\times P}$ are respectively the capacitance, conductivity, and advection matrices for element i. The advection matrix depends on $\mathbf{v}_m$ and thus Eq. (55) is a non-linear function of $\mathbf{u}_i$. Since the trial functions are orthogonal and ρc_p is constant within an element, $\mathbf{A}^i$ is diagonal and positive definite as $\rho c_p > 0$.

For compactness, the element-wise model in Eq. (55) is also written in matrix form,

$$\mathbf{A}\dot{\mathbf{u}} = \left[\mathbf{B} + \mathbf{C}(\mathbf{u})\right]\mathbf{u} + \mathbf{f}(t) \tag{57}$$

where $\mathbf{u} = \left[\mathbf{u}^{(1)}, \mathbf{u}^{(2)}, \cdots, \mathbf{u}^{(M)}\right]^T \in \mathbb{R}^{MP}$ includes all DG variables, $\mathbf{f} = \left[\mathbf{f}^{(1)}, \mathbf{f}^{(2)}, \ldots, \mathbf{f}^{(M)}\right]^T \in \mathbb{R}^{MP}$, $\mathbf{A}$ and $\mathbf{C}$ are matrices of M diagonal blocks whose i-th blocks are $\mathbf{A}^i$ and $\mathbf{C}^i$, and $\mathbf{B}$ is a matrix of $M \times M$ blocks whose (i, j)-th block is,

$$\mathbf{B}_{ij} = \begin{cases} \mathbf{B}^i + \sum_{j\in\mathcal{N}_i\cup\{T_b\}} \mathbf{B}_{ij}^i, & i = j \\ \mathbf{B}_{ij}^j, & i \neq j \end{cases} \tag{58}$$

The dependency of $\mathbf{C}$ on $\mathbf{u}$ is explicitly noted in Eq. (57), which is the main source of non-linearity in the current TPS problem. Moreover, the mesh velocity $\mathbf{v}$ varies with space and time, and thus the advection matrix $\mathbf{C}$ varies with time as a function of the surface temperature $T_q(x,t)$.

B. Coarse-Graining of Dynamics

The LCM is obtained by coarse-graining the full-order DG-FEM. This coarse-graining procedure produces resolved $\mathbf{r}^{(1)}(\bar{\mathbf{u}},t)$ and residual $\mathbf{r}^{(2)}(\mathbf{u},t)$ dynamics as in Eq. (24). This section presents the detail derivations and magnitude analysis for the resolved and residual dynamics.

1. Resolved Dynamics

Using Eq. (21), the resolved dynamics is computed as follows,

$$\mathbf{r}^{(1)}(\bar{\mathbf{u}},t) = \mathcal{P}\left[\mathbf{\Phi}^+\mathbf{A}^{-1}\left(\mathbf{B}\mathbf{u} + \mathbf{C}(\mathbf{u})\mathbf{u} + \mathbf{f}(t)\right)\right] \tag{59a}$$

$$= \mathbf{\Phi}^+\mathbf{A}^{-1}\mathbf{P}\mathbf{B}\mathbf{P}\mathbf{u} \quad + \mathbf{\Phi}^+\mathbf{A}^{-1}\mathbf{P}\mathbf{C}\left(\mathbf{P}\mathbf{u}\right)\mathbf{P}\mathbf{u} + \mathbf{\Phi}^+\mathbf{A}^{-1}\mathbf{P}\mathbf{f}(t) \tag{59b}$$

$$= \underbrace{\mathbf{\Phi}^+\mathbf{A}^{-1}\mathbf{\Phi}}_{\#1}\underbrace{\mathbf{\Phi}^+\mathbf{B}\mathbf{\Phi}\bar{\mathbf{u}}}_{\#2} \quad + \mathbf{\Phi}^+\mathbf{A}^{-1}\mathbf{\Phi}\underbrace{\mathbf{\Phi}^+\mathbf{C}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi}}_{\#3}\bar{\mathbf{u}} + \mathbf{\Phi}^+\mathbf{A}^{-1}\mathbf{\Phi}\underbrace{\mathbf{\Phi}^+\mathbf{f}(t)}_{\#4} \tag{59c}$$

Detailed derivations for the #1, #2, and #4 terms can be found in Ref. [10] for the case of temperature-varying matrices, where it is shown that coarse-graining the capacitance, conductivity, and forcing terms result exactly in the LCM matrices. The remaining term #3 is analyzed next.

Term #3 The $\mathbf{C}(\mathbf{u}) \in \mathbb{R}^{MP\times MP}$ matrix id block diagonal with M matrices of size $P\times P$, since the basis functions are defined locally on each element. Therefore, $[\mathbf{C}(\mathbf{u})]_{ij} = \mathbf{0}$ for all $i \neq j$ with $i,j = 1,2,\ldots,M$. It follows that for $k,l = 1,2,\cdots,P$,

$$\left[\mathbf{\Phi}^+\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\mathbf{\Phi}\right]_{kl} = \sum_{i=1}^{M}\sum_{j=1}^{M}\boldsymbol{\varphi}_i^{k+}\left[\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\right]_{ij}\boldsymbol{\varphi}_j^l \tag{60a}$$

$$= \sum_{i=1}^{M}\boldsymbol{\varphi}_i^{k+}\left[\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\right]_{ii}\boldsymbol{\varphi}_i^l \tag{60b}$$

$$= \sum_{i\in\mathcal{V}_k}\boldsymbol{\varphi}_i^{k+}\left[\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\right]_{ii}\boldsymbol{\varphi}_i^l \tag{60c}$$

where in the second row, the fact that $[\mathbf{C}(\mathbf{u})]_{ij} = 0$ for all $i \neq j$ is used, and in the last row, the fact that $\boldsymbol{\varphi}_i^{k+} = 0$ for all $i \notin \mathcal{V}_k$ is used. Now, considering that $[\mathbf{C}(\mathbf{u})]_{ii}$ has a $(1,1)$-th zero element, i.e., $[C_{11}(\mathbf{\Phi}\bar{\mathbf{u}})]_{ii} = 0$, and that if $k \neq l$ then $i \notin \mathcal{V}_l$ and thus $\boldsymbol{\varphi}_i^l = \mathbf{0}$, it follows that for some index $i \in \mathcal{V}_k$,

$$\boldsymbol{\varphi}_i^{k+}\left[\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\right]_{ii}\boldsymbol{\varphi}_i^l = \boldsymbol{\varphi}_i^{k+}\left[\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\right]_{ii}\boldsymbol{\varphi}_i^k = \frac{|E_i|}{|\Omega_k|}\left[C_{11}(\mathbf{\Phi}\bar{\mathbf{u}})\right]_{ii} = 0 \tag{61}$$

The matrix $[\mathbf{\Phi}^+\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\mathbf{\Phi}]_{kl} = 0$ for all $k,l = 1,2,\ldots,P$, and thus,

$$\bar{\mathbf{C}}\left(\bar{\mathbf{u}}\right) = \mathbf{\Phi}^+\mathbf{C}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi} = \mathbf{0} \tag{62}$$

as indicated by the LCM in Eq. (11).

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2. Magnitude Analysis for Residual Dynamics

Next, the magnitude of the residual dynamics $\mathbf{r}^{(2)}(\mathbf{u}, t)$ is analyzed to pinpoint the missing physics in the LCM. By definition,

$$\mathbf{r}^{(2)}(\mathbf{u}, t) = \dot{\bar{\mathbf{u}}} - \mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) \tag{63a}$$

$$= \mathbf{\Phi}^{+}\mathbf{r}(\mathbf{u}, t) - \mathbf{r}^{(1)}(\bar{\mathbf{u}}, t) \tag{63b}$$

$$= \underbrace{\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{B}\mathbf{u} - \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{B}}(\mathbf{w})\bar{\mathbf{u}}}_{\#1} + \underbrace{\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{C}}(\bar{\mathbf{u}})\bar{\mathbf{u}}}_{\#2} + \underbrace{\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{f}(t) - \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{f}}(t)}_{\#3} \tag{63c}$$

The magnitude analysis for terms #1 and #3 can be found in Ref. [10] for temperature-varying material properties. The remaining term #2 is analyzed next. Let $\mathbf{D}(\bar{\mathbf{u}}) = \mathbf{A}^{-1}\mathbf{P}\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})$ so that,

$$\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{C}}(\bar{\mathbf{u}})\bar{\mathbf{u}} \tag{64a}$$

$$= \mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{P}\mathbf{C}(\mathbf{\Phi}\bar{\mathbf{u}})\mathbf{\Phi}\mathbf{\Phi}^{+}\mathbf{u} \tag{64b}$$

$$= \mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \mathbf{\Phi}^{+}\mathbf{D}(\bar{\mathbf{u}})\mathbf{\Phi}\mathbf{\Phi}^{+}\mathbf{u} \tag{64c}$$

$$\tag{64d}$$

where $\mathbf{P} = \mathbf{\Phi}\mathbf{\Phi}^{+}$. Thus,

$$\left\|\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \bar{\mathbf{A}}(\mathbf{w})^{-1}\bar{\mathbf{C}}(\bar{\mathbf{u}})\bar{\mathbf{u}}\right\| \tag{65a}$$

$$\leq \left\|\mathbf{\Phi}^{+}\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \mathbf{\Phi}^{+}\mathbf{D}(\bar{\mathbf{u}})\mathbf{u}\right\| + \left\|\mathbf{\Phi}^{+}\mathbf{D}(\bar{\mathbf{u}})\mathbf{u} - \mathbf{\Phi}^{+}\mathbf{D}(\bar{\mathbf{u}})\mathbf{\Phi}\mathbf{\Phi}^{+}\mathbf{u}\right\| \tag{65b}$$

$$\leq \left\|\mathbf{\Phi}^{+}\right\| \underbrace{\left\|\mathbf{A}^{-1}\mathbf{C}(\mathbf{u})\mathbf{u} - \mathbf{D}(\bar{\mathbf{u}})\mathbf{u}\right\|}_{\#1} + \left\|\mathbf{\Phi}^{+}\mathbf{D}(\bar{\mathbf{u}})\right\| \underbrace{\left\|\mathbf{u} - \mathbf{\Phi}\mathbf{\Phi}^{+}\mathbf{u}\right\|}_{\#2} \tag{65c}$$

Similar to the analysis in Ref. [10], term #2 is due to the approximation of non-uniform temperature as constants in each component, and the term #1 is the error in the advection effects due to such approximation.

C. Lumped Capacitance Model

The following assumptions are employed: (1) the temperature in the i-component is described by a scalar time-varying average temperature $\bar{u}_i$, (2) between neighboring components (i) and (j) the heat flux is approximated as,

$$q_{ij} = \frac{\bar{u}_j - \bar{u}_i}{R_{ij}} \tag{66}$$

where R_{ij} is the thermal resistance. Empirically, for a component of isotropic heat conductivity k, length ℓ, and cross-section area A, the thermal resistance is $R = \ell / kA$. Between components i and j, define $R_{ij} = R_i + R_j$. In addition, the heat flux due to Dirichlet boundary condition is computed as $q_{iT} = (T_b - \bar{u}_i)/R_i$.

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At component i, the dynamics of LCM are given by,

$$\int_{E_i} \rho c_p \dot{\bar{u}}_i dE_i = \left(\sum_{j \in \mathcal{N}_i} \int_{e_{ij}} \frac{\bar{u}_j - \bar{u}_i}{R_{ij}} de_{ij} \right) + \int_{e_{iq}} q_b de_{iq} + \int_{e_{iT}} \frac{T_b - \bar{u}_i}{R_i} de_{iT} \tag{67a}$$

$$\bar{A}_i \dot{\bar{u}}_i = \left(\sum_{j \in \mathcal{N}_i} \frac{|e_{ij}|}{R_{ij}} \left(\bar{u}_j - \bar{u}_i\right) \right) + |e_{iq}| \bar{q}_i + \frac{|e_{iT}|}{R_i} \left(\bar{T}_i - \bar{u}_i\right) \tag{67b}$$

$$= \sum_{j \in \mathcal{N}_i} \left(-\frac{|e_{ij}|}{R_{ij}} \bar{u}_i + \frac{|e_{ij}|}{R_{ij}} \bar{u}_j \right) + \left(-\frac{|e_{iT}|}{R_i} \bar{u}_i \right) + \left(|e_{iq}| \bar{q}_i + \frac{|e_{iT}|}{R_i} \bar{T}_i \right) \tag{67c}$$

$$= \sum_{j \in \mathcal{N}_i \cup \{T_b\}} \left(\bar{B}^i_{ij} \bar{u}_i + \bar{B}^j_{ij} \bar{u}_j \right) + \bar{f}_i \tag{67d}$$

where in Eq. (67b) $|e|$ denotes the length ($d = 2$) or area ($d = 3$) of a component boundary e. The $\bar{A}_i$, $\bar{B}^i_{ij}$, and $\bar{B}^j_{ij}$ quantities are provided in Eq. (14).

The lumped-mass representation for the four-component TPS is shown in Fig. 3. With $\mathbf{w} \in \mathbb{R}^{\tilde{N}}$ as the one-dimensional surface displacements, v_i represent the area of the i-th component, $\overline{\rho c_p}_{,i}$, the constant heat capacitance, and $1/R_{ij} = 1/R_i + 1/R_j$ the equivalent thermal resistance between i and j. Note that the areas and equivalent thermal resistances vary with $\mathbf{w}$ due to ablation. Leveraging the formulas from Eqs. (13b) and (14), the LCM matrices are given by,

$$\bar{\mathbf{A}} = \begin{bmatrix} \overline{\rho c_p}_{,1} v_1 & 0 & 0 & 0 \\ 0 & \overline{\rho c_p}_{,2} v_2 & 0 & 0 \\ 0 & 0 & \overline{\rho c_p}_{,3} v_3 & 0 \\ 0 & 0 & 0 & \overline{\rho c_p}_{,4} v_4 \end{bmatrix}, \tag{68a}$$

$$\bar{\mathbf{B}} = \begin{bmatrix} \frac{1}{R_{12}} + \frac{1}{R_{14}} & -\frac{1}{R_{12}} & 0 & -\frac{1}{R_{14}} \\ -\frac{1}{R_{12}} & \frac{1}{R_{12}} + \frac{1}{R_{24}} + \frac{1}{R_{23}} & -\frac{1}{R_{23}} & -\frac{1}{R_{24}} \\ 0 & -\frac{1}{R_{32}} & \frac{1}{R_{32}} + \frac{1}{R_{34}} & -\frac{1}{R_{34}} \\ -\frac{1}{R_{14}} & -\frac{1}{R_{24}} & -\frac{1}{R_{34}} & \frac{1}{R_{14}} + \frac{1}{R_{24}} + \frac{1}{R_{34}} \end{bmatrix}, \quad \bar{\mathbf{f}} = \begin{bmatrix} \bar{q}^{(1)} \\ \bar{q}^{(2)} \\ \bar{q}^{(3)} \\ 0 \end{bmatrix} \tag{68b}$$

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References

[1] Amar, A. J., Oliver, A. B., Kirk, B. S., Salazar, G., and Droba, J., "Overview of the CHarring Ablator Response (CHAR) Code," *AIAA Aviation Forum*, edited by AIAA, 2016. https://doi.org/10.2514/6.2016-3385.

[2] Howard, M. A., and Blackwell, B. F., "A Multi-Dimensional Finite Element Based Solver for Decomposing and Non-decomposing Thermal Protection Systems," *AIAA Aviation Forum*, edited by AIAA, 2015. https://doi.org/10.2514/6.1015-2506.

Downloaded by Sandia National Laboratories on June 15, 2026 | http://arc.aiaa.org | DOI: 10.2514/6.2026-1207

[3] Cohen, G., and Pernet, S., *Finite Element and Discontinuous Galerkin Methods for Transient Wave Equations*, Springer Dordrecht, Le Chesnay and Toulouse France, 2018. https://doi.org/10.1007/978-94-017-7761-2.

[4] Fang, J., Hu, W., Liu, Z., Zhou, Y., Wei, C., and Tan, J., "A Reduced Order Finite Element-Informed Surrogate Model for Approximating Global High-Fidelity Simulations," *Structural and Multidisciplinary Optimization*, Vol. 67, No. 12, 2024. https://doi.org/10.1007/s00158-024-03935-3.

[5] Neelan, A. G., "Linear and Non-linear Reduced Order Model of Scramjet," *Advances in Multidisciplinary Design, Analysis and Optimization*, Singapore, 2025, pp. 111–118. https://doi.org/10.1007/978-981-96-1158-4_13.

[6] Xian, Y., Pan, B., and Luo, L., "A New Model Updating Strategy with Physics-Based and Data-Driven Models," *Structural and Multidisciplinary Optimization*, Vol. 64, No. 1, 2021. https://doi.org/10.1007/s00158-021-02868-5.

[7] Rasmussen, C. E., and Williams, C. K. I., *Gaussian Processes for Machine Learning*, The MIT Press, 2006.

[8] Chen, R. T. Q., Rubanova, Y., Bettencourt, J., and Duvenaud, D., "Neural Ordinary Differential Equations," , 2019. https://doi.org/10.48550/arXiv.1806.07366.

[9] Thelen, A., Zhang, X., Fink, O., Lu, Y., Ghosh, S., Youn, B. D., Todd, M. D., Mahadean, S., Hu, C., and Hu, Z., "A comprehensive review of digital twin – part 1: modeling and twinning enabling technologies," *Structural and Multidisciplinary Optimization*, Vol. 65, No. 12, 2022. https://doi.org/10.1007/s00158-022-03425-4.

[10] Venegas, C. A. V., Huang, D., Blonigan, P., and JohnTencer, "Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Hypersonic Thermal Protection Systems," , 2025. https://doi.org/10.48550/arXiv.2505.22890.

[11] Parish, E., and Duraisamy, K., *Coarse-Graining Turbulence Using the Mori–Zwanzig Formalism*, Cambridge University Press, Cambridge, 2025.

[12] Parish, E., and Duraisamy, K., "A Unified Framework for Multiscale Modeling using the Mori-Zwanzig Formalism and the Variational Multiscale Method," *ArXiv*, 2017. https://doi.org/10.48550/arXiv.1712.09669.

[13] Parish, E., and Duraisamy, K., "Non-Markovian Closure Models for Large Eddy Simulations Using the Mori-Zwanzig Formalism," *Phys. Rev. Fluids*, Vol. 2, 2017, p. 014604. https://doi.org/10.1103/PhysRevFluids.2.014604.

[14] Clausen, J., Brunini, V., Collins, L., Knaus, R. C., Kucala, A., Lin, S., Moser, D. R., Phillips, M., Ransegnola, T. M., Subia, S. R., et al., "SIERRA Multimechanics Module: Aria Verification Manual-Version 5.22," Tech. rep., Sandia National Lab.(SNL-NM), Albuquerque, NM (United States), 2024.

[15] Incropera, F. P., DeWitt, D. P., Bergman, T. L., and Lavine, A. S., *Fundamentals of Heat and Mass Transfer*, 7$^{\text{th}}$ ed., Wiley, Hoboken, NJ, 2011.

[16] Yu, Y., Harlim, J., Huang, D., and Li, Y., "Learning Coarse-Grained Dynamics on Graph," *arXiv preprint arXiv:2405.09324*, 2024. https://doi.org/10.48550/arXiv.2405.09324.

[17] Padovan, A., Vollmer, B., Panerai, F., Panesi, M., Stephani, K. A., and Bodony, D. J., "An Extended BF́ormulation for Ablating-surface Boundary Conditions," *International Journal of Heat and Mass Transfer*, Vol. 218, 2024, p. 124770. https://doi.org/10.1016/j.ijheatmasstransfer.2023.124770.

[18] Yu, Y., Huang, D., Park, S., and Pangborn, H., "Learning Networked Dynamical System Models with Weak Form and Graph Neural Networks," *arXiv preprint arXiv:2407.16779*, 2024. https://doi.org/10.48550/arXiv.2407.16779.